

Muscular-Skeletal Systems and Locomotion



Key Concepts:

- **Types of Animal Skeletons**
- **Skeletal Muscle Structure and the Mechanism of Force Generation**
- **Types of Skeletal Muscle Fibers and Their Functions**
- **Muscle remodelling**
- **Impact on Public Health**

Types of Animal Skeletons

Skeleton's functions:

support, protection, and locomotion (the movement from place to place)

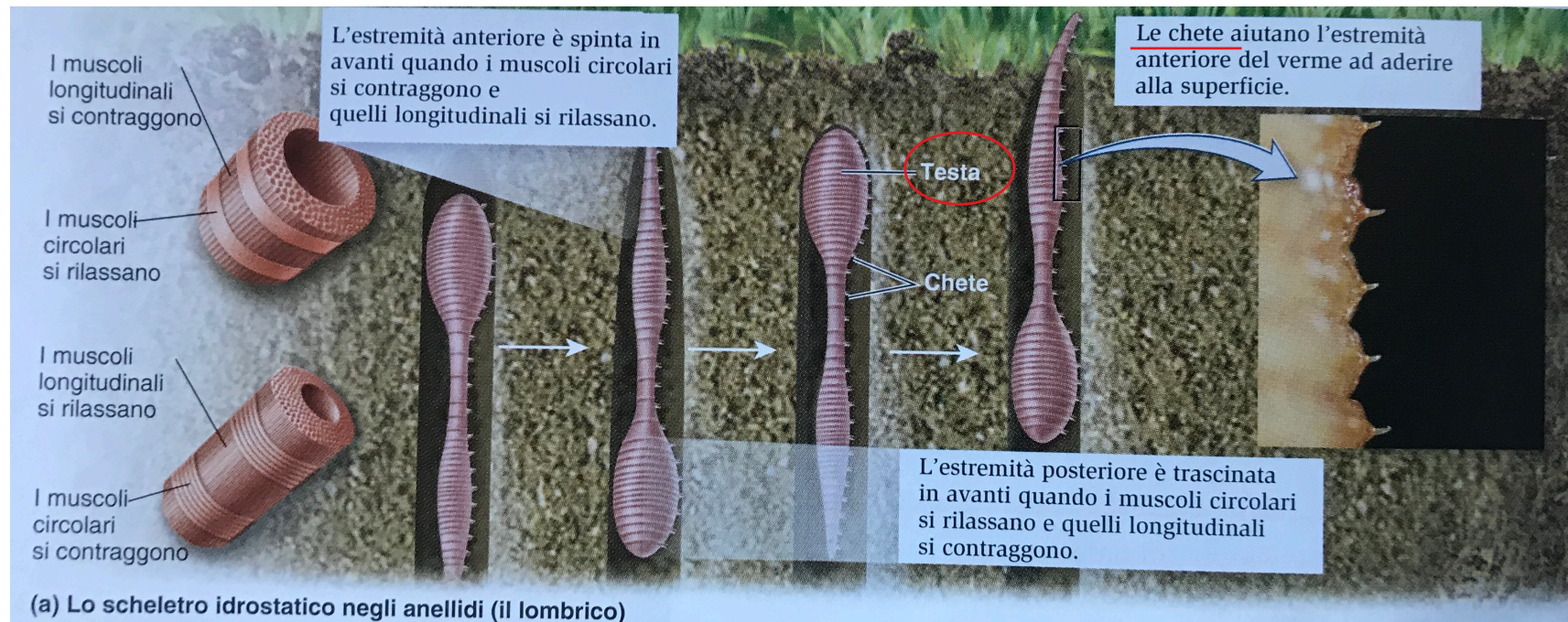
Three types of skeletons

- **Hydroskeleton**
- **Exoskeleton**
- **Endoskeleton**

Hydroskeletons

Hydroskeleton → body compartments filled with liquid

- Found in some **soft-bodied invertebrates** that use **water pressure for propulsion**
- **Longitudinal and circular muscles** generate contraction waves that allows movements

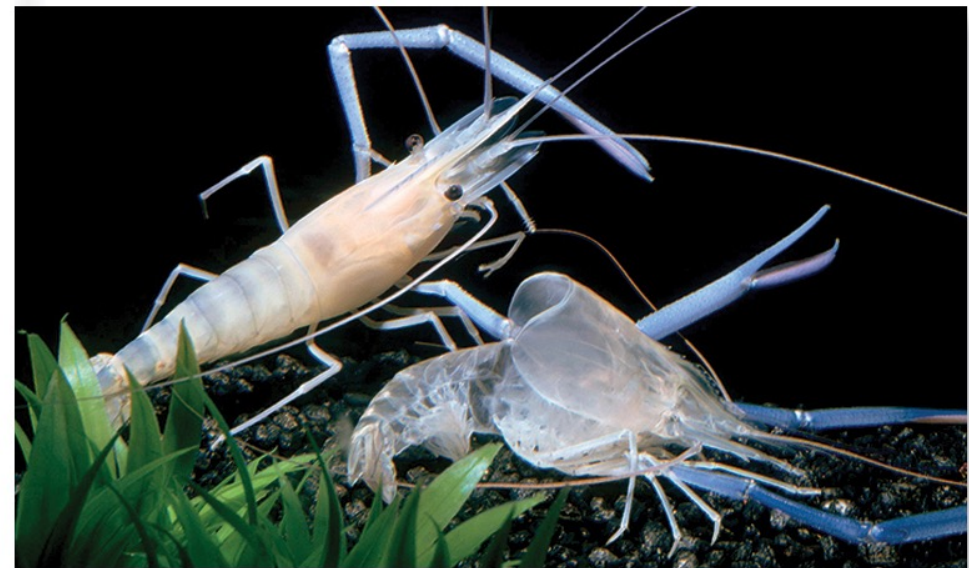
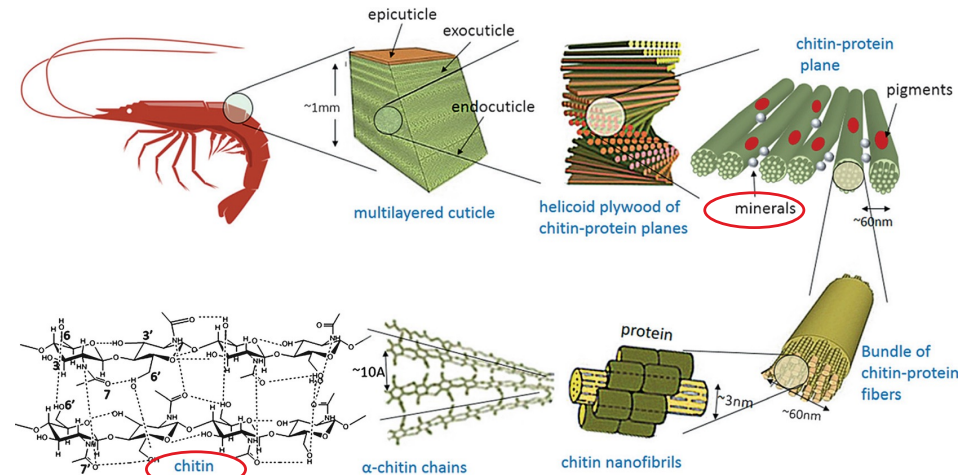


Exoskeletons

External skeleton that **surrounds** and **protects body** surface

Arthropod exoskeleton

- Made of the **polysaccharide chitin**
 - Sometimes strengthened with minerals such as **calcium**
 - Immersed in a **protein matrix**
- **Tough and durable**
- Vary in complexity, thickness, and durability
- **Segmented** for movement
- Growth requires **molting**
 - vulnerability to predators during molting

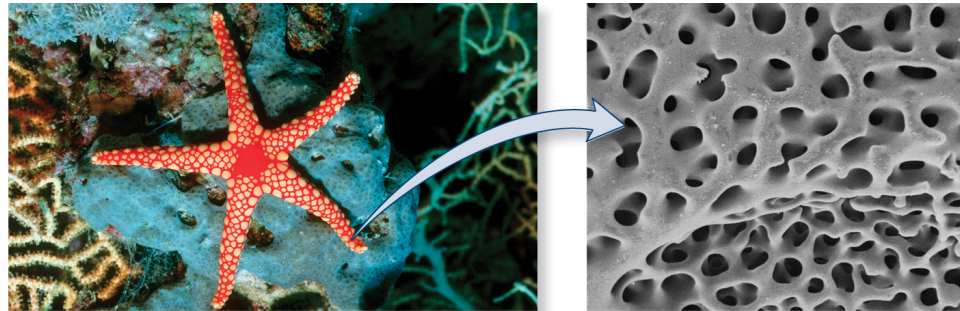


a) An arthropod next to its recently shed exoskeleton

Endoskeletons

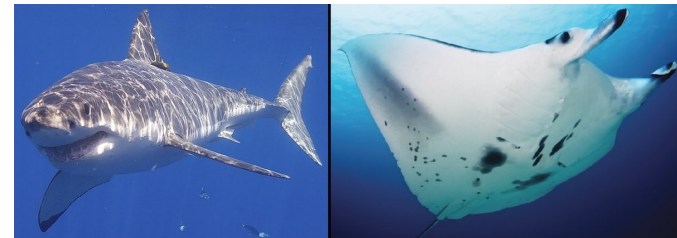
- **Internal structures**
- Do **not protect body surface**, only some internal organs (i.e., skull)
- Found in **sponges, echinoderms, and vertebrates**
- **Minerals** provide firmness
- **Sponge and echinoderm** skeletons composed of spiky **networks of proteins and minerals** or mineralized plate-like structures

b) Echinoderm (sea star) and (right)
SEM of its endoskeleton

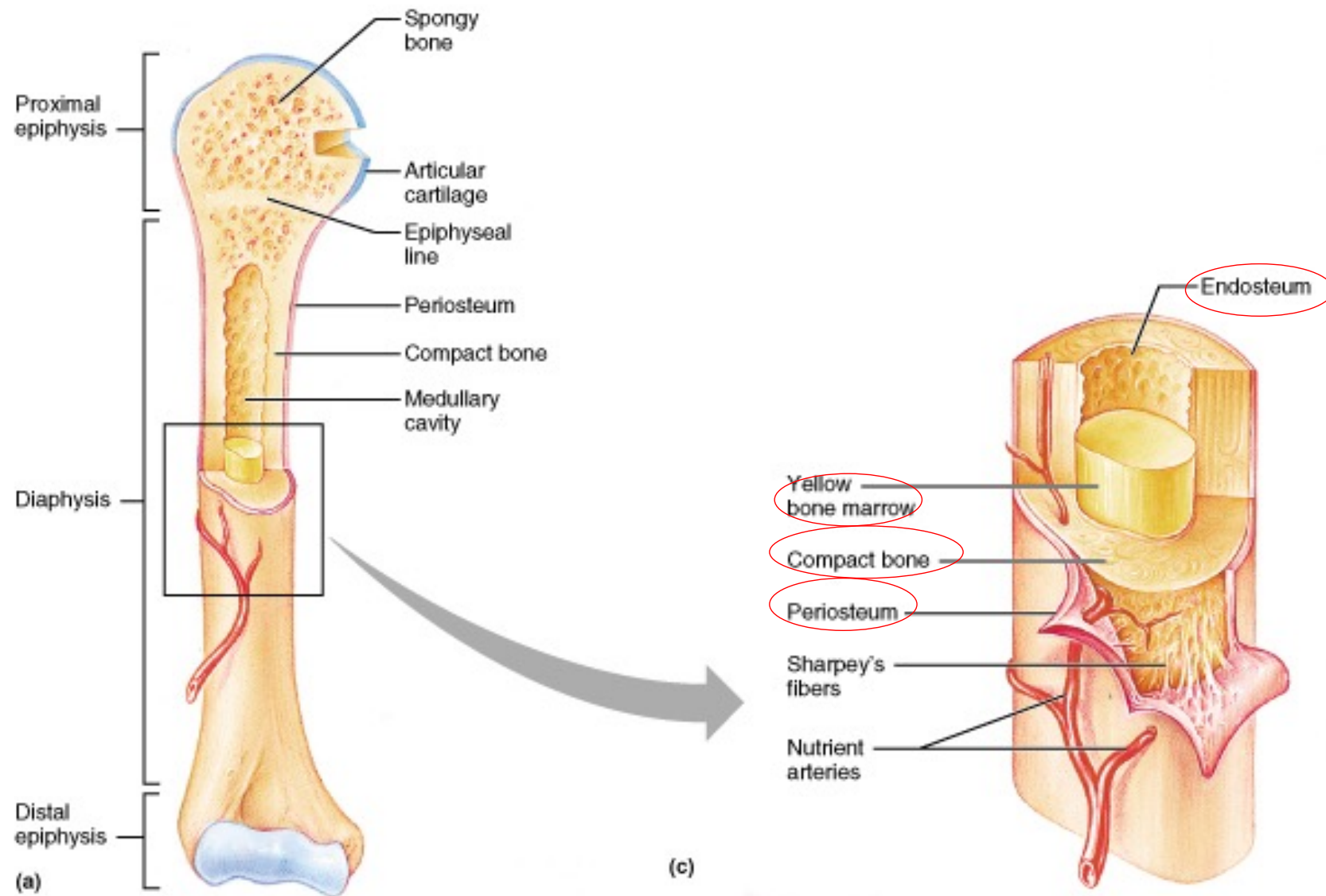


- **Vertebrate** skeletons composed:

- entirely of **cartilage** (cartilaginous fishes)
- **cartilage and bone** (bony fishes, amphibians, reptiles, birds, mammals)



Bone structure



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Bone

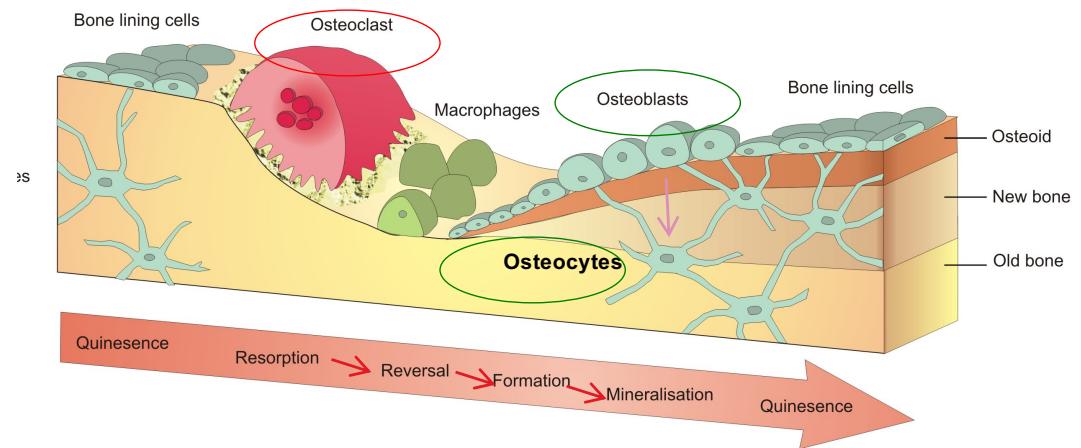
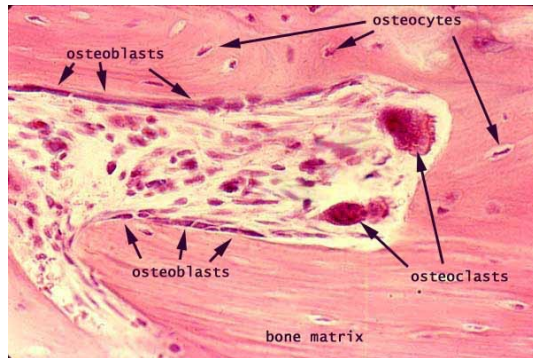
Mineral component

- Crystalline mixture of Ca^{2+} , PO_4^- and other ions – provide rigidity

Organic components

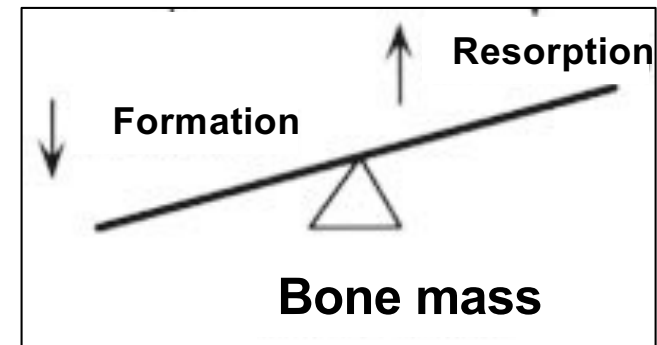
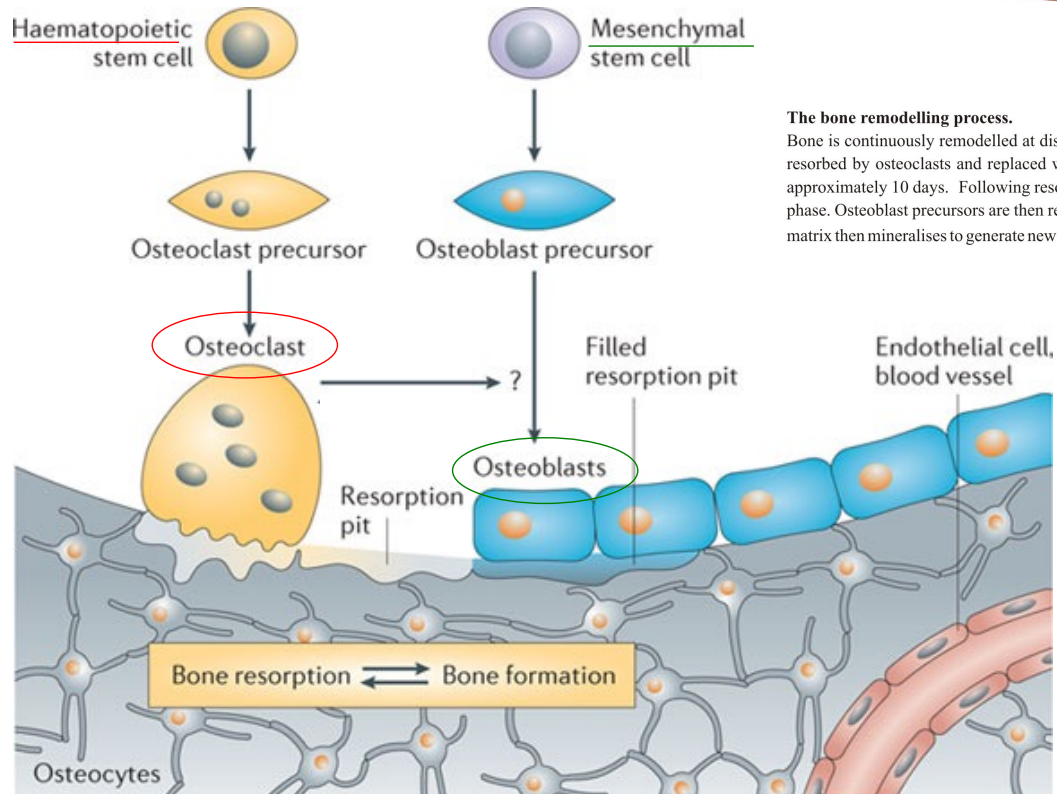
- **Osteoblasts and osteocytes** – cells that form bone
 - secrete collagen – triple helical structure provides strength and flexibility
 - **Osteoclasts** break down bone
- Living, dynamic tissue

Bone remodeling: the actors



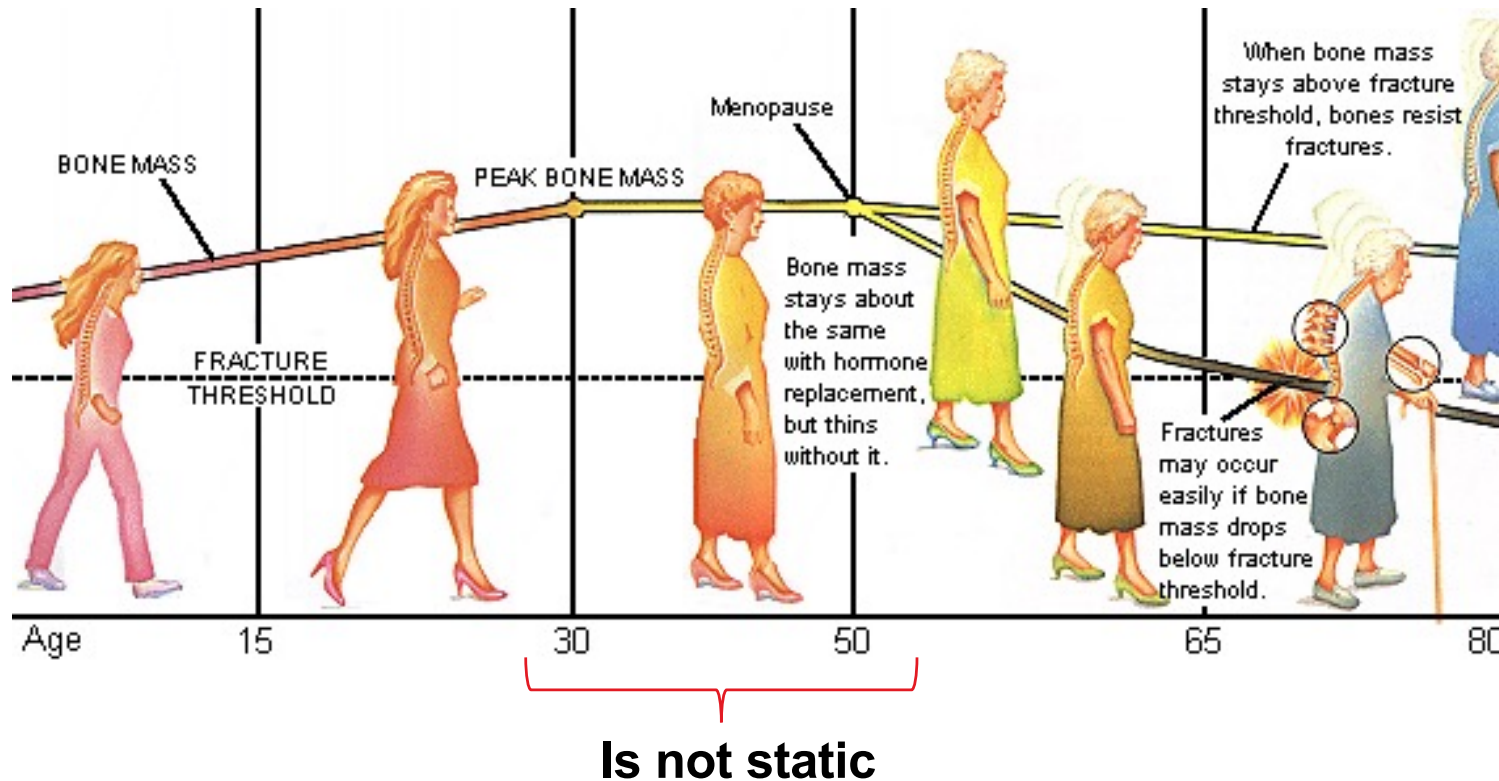
The bone remodelling process.

Bone is continuously remodelled at discrete sites in the skeleton in order to maintain the integrity of the tissue. During this process, old bone is resorbed by osteoclasts and replaced with new osteoid, secreted by osteoblasts. First osteoclasts are activated, and the resorption phase takes approximately 10 days. Following resorption, unclassified macrophage-like cells are found at the remodelling site in the intermediate, or reversal phase. Osteoblast precursors are then recruited, which proliferate and differentiate into mature osteoblasts, before secreting new bone matrix. The matrix then mineralises to generate new bone and this completes the remodelling process. Copyright BTR©



Bone remodeling

- Bone mass is changing during life



- Bones are continuously remodeling** at discrete sites also in absence of injuries

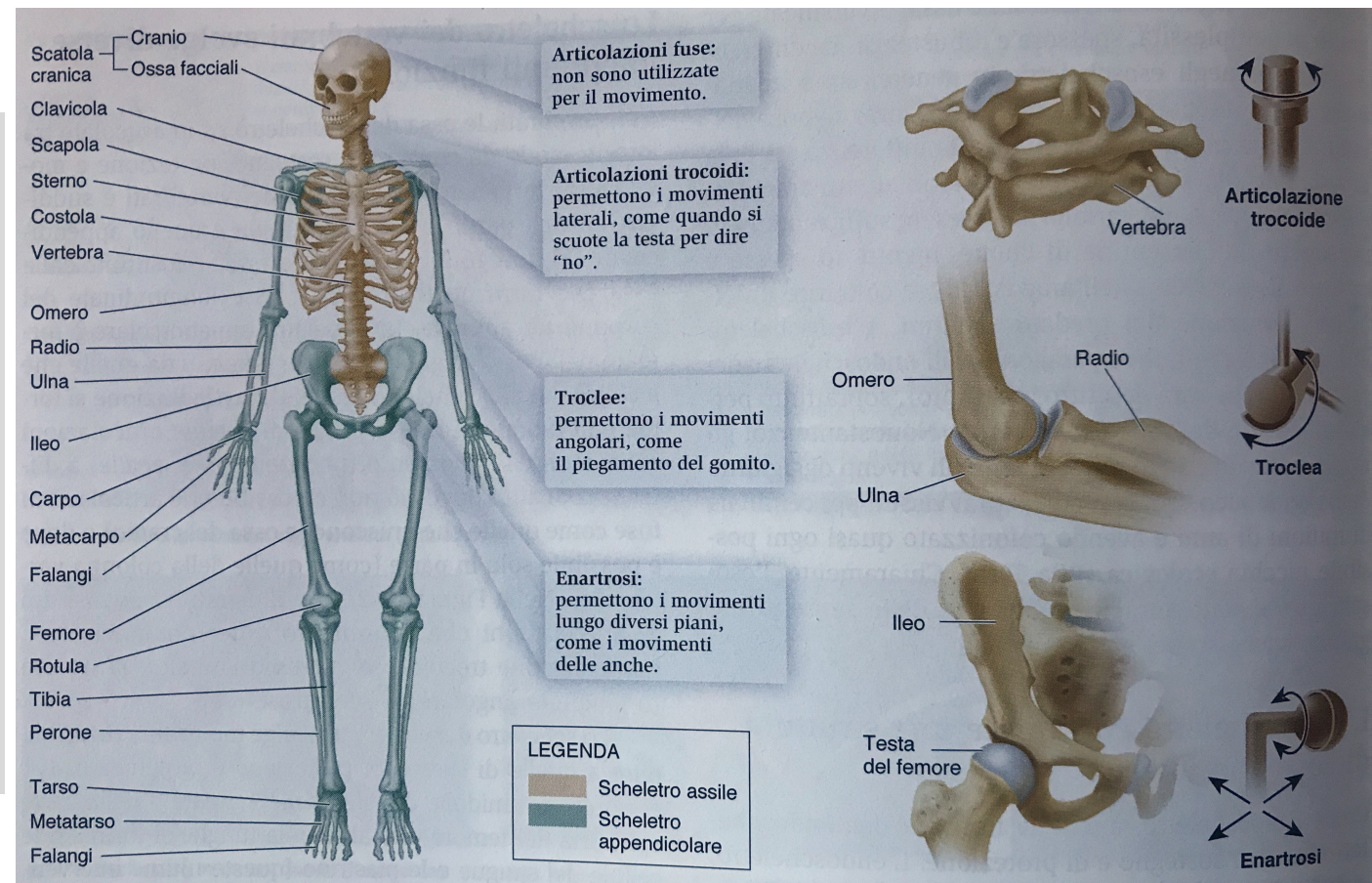
The vertebrate skeleton

Two parts

- **Axial skeleton** – main longitudinal axis
- **Appendicular skeleton** – limb bones and girdles

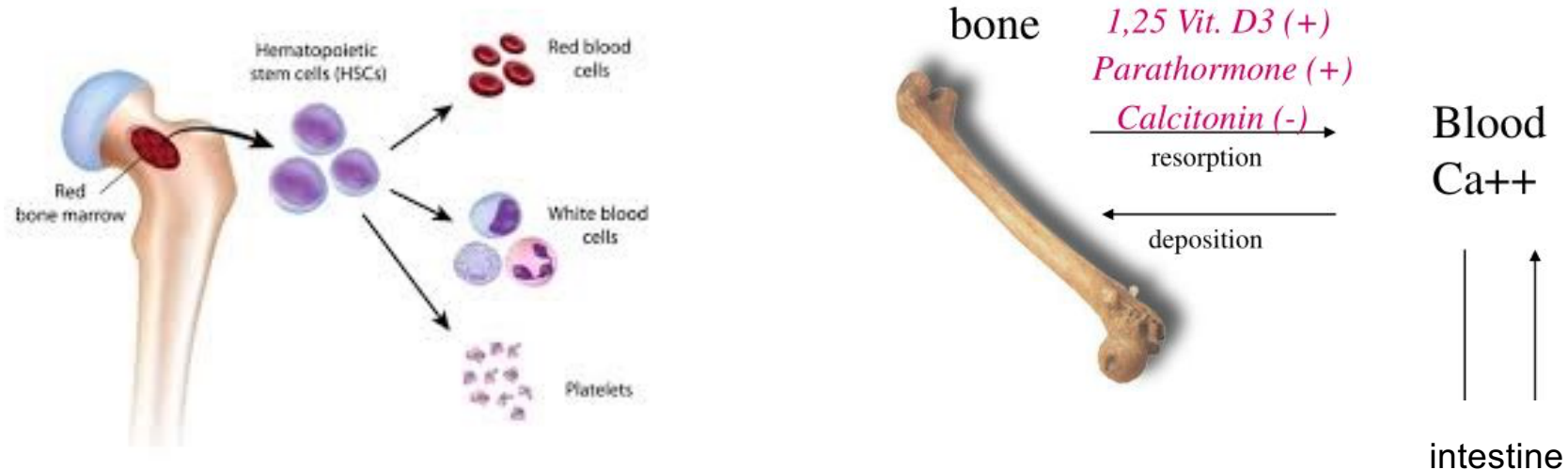
Joint – formed where two or more bones come together

- Three types that allow movement
 - **Pivot joints**
 - **Hinge joints**
 - **Ball-and-socket joint**
- Others do not
 - **Fused joints** (i.e. skull)



Additional functions of the skeleton

- **Blood cells and platelets** are formed in **bone marrow**
- Ca^{2+} and PO_4^{2-} homeostasis achieved in large part through **exchanges between bone and blood**



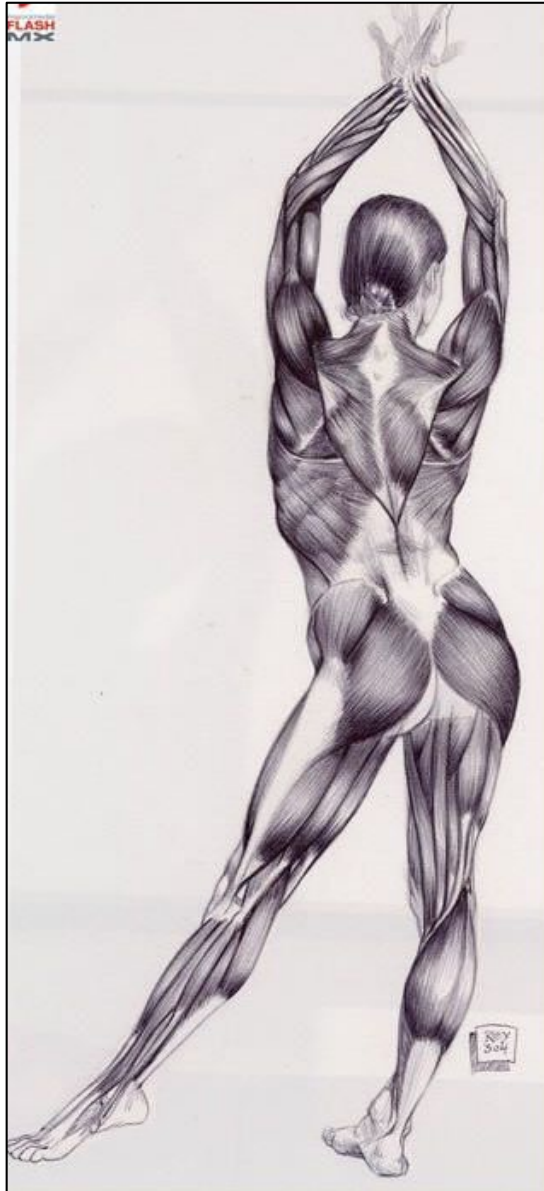
Skeletal Muscle Structure and the Mechanism of Force Generation

Vertebrates have **three types of muscle**, classified according to structure, function, and control mechanisms

- **Smooth muscle** – surrounds and forms part of the lining of hollow organs and tubes
- **Cardiac muscle** – found only in the heart
- **Skeletal muscle** – directly involved in locomotion
 - Electrically excitable in vertebrates

} striated

Skeletal muscle



Movement/posture



Metabolism



Heat



Other (i.e. breathing)

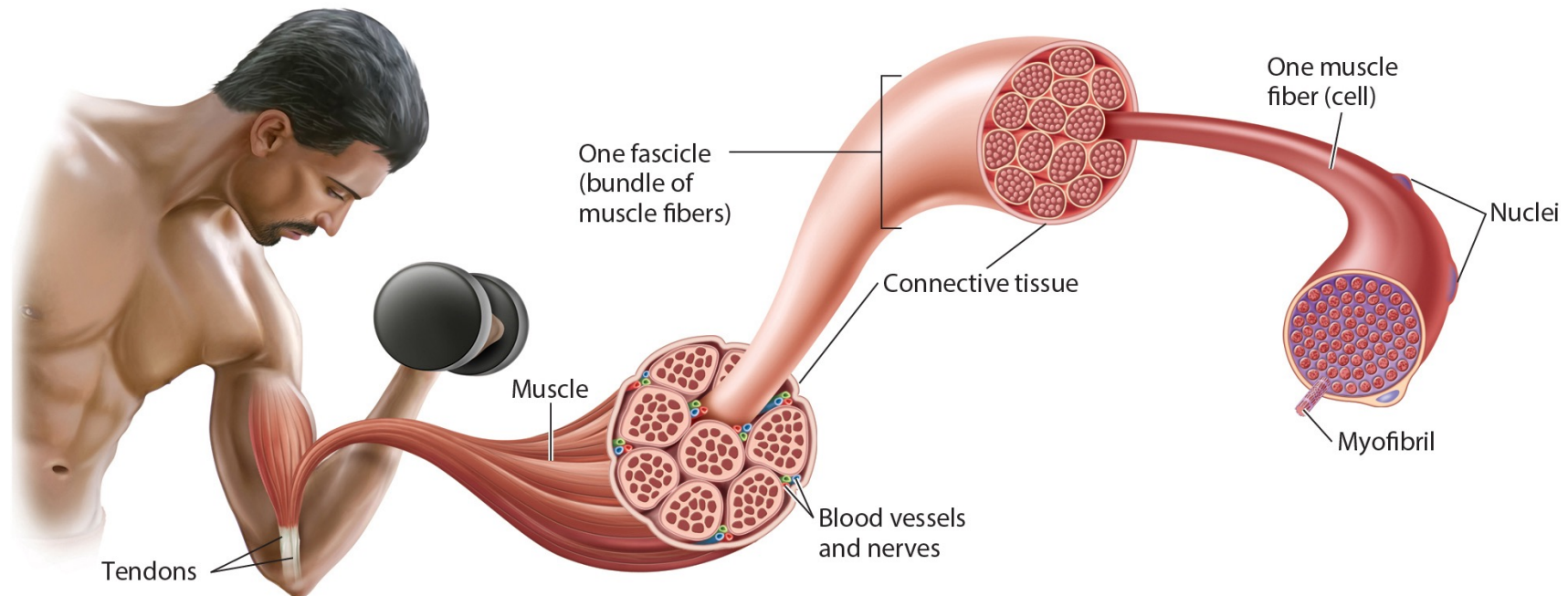


Skeletal muscle

A grouping of cells (**muscle fibers**) bound together in **bundles (fascicles)** by connective tissue

Usually linked to bones by **bundles of collagen fibers** called **tendons**

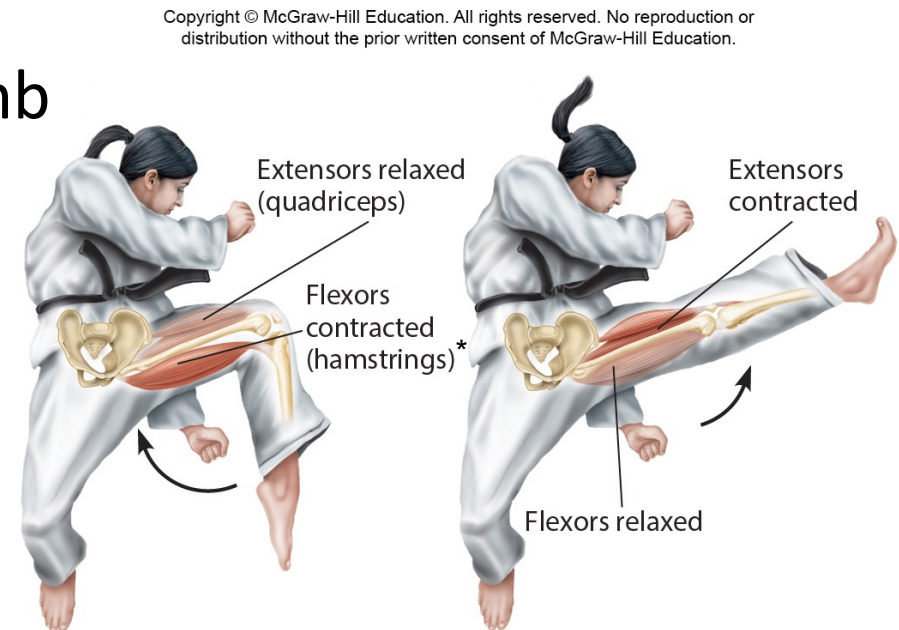
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Types of skeletal muscles

Bone moves as muscle shortens

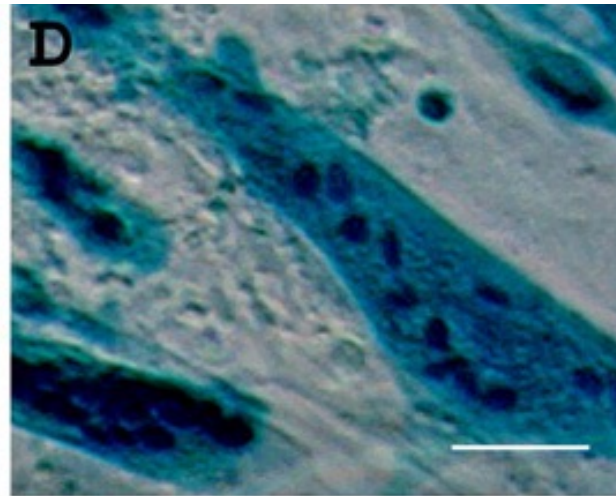
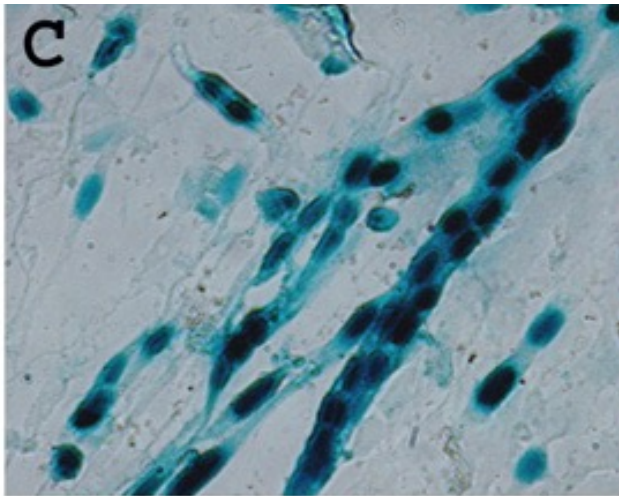
- **Contracting muscle exerts only a pulling force** – attached bones pulled toward or away from each other:
 - **Flexors** bend a limb at a joint
 - **Extensors** straighten a limb
 - Groups of muscles that produce **oppositely directed movements** at a joint are **antagonists**



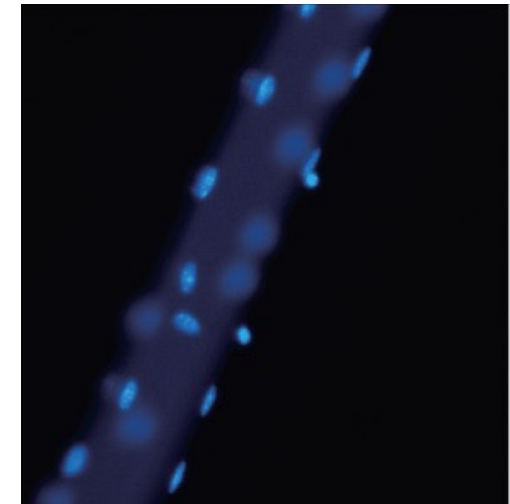
*Hamstring= termine inglese per definire i muscoli flessori della coscia, ossia il bicipite femorale, il semimembranoso e il semitendinoso

Muscle fibers are syncytia

- Skeletal muscle fibers arise from **cells that fuse** to form a single cell with multiple nuclei → **Syncytium**



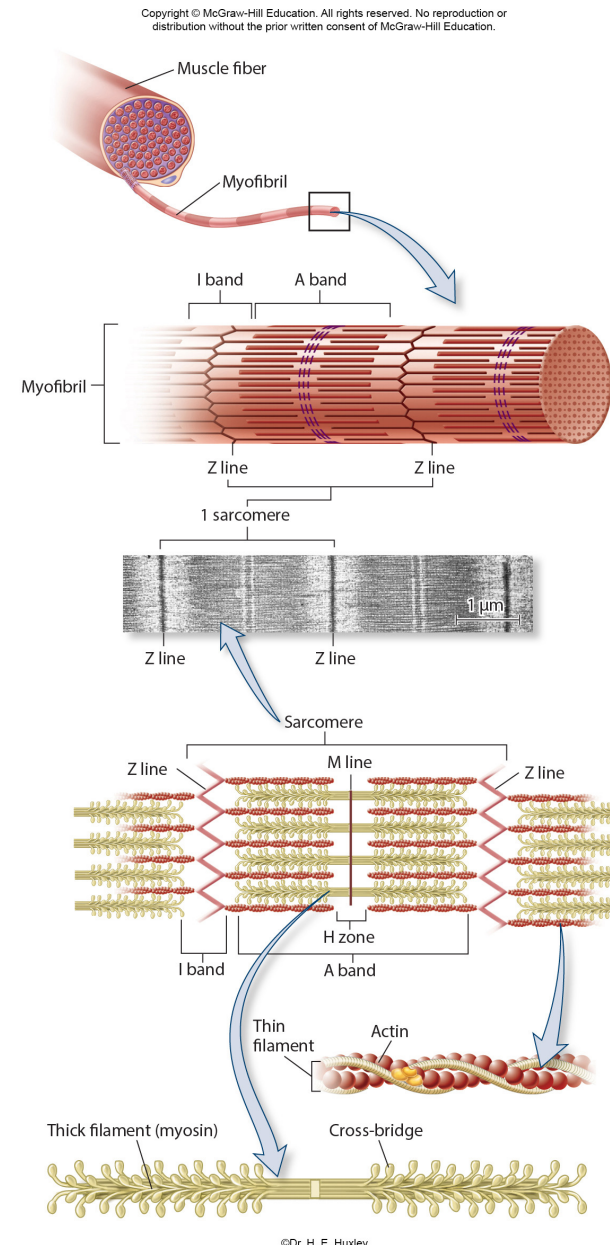
In vitro



Ex vivo

Muscle fibers: contracting apparatus

- **Myofibrils** – cylindrical bundles in muscle fibers
- Each myofibril contains repeating units of filaments (**sarcomeres**)
 - **Thick filaments** – made of **myosin**
 - **Thin filaments** – made of **actin** and other proteins



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Sarcomeres

Z line – network of proteins to which thin filaments are attached

- Two successive Z lines define a sarcomere

A band – wide formed by thick filaments

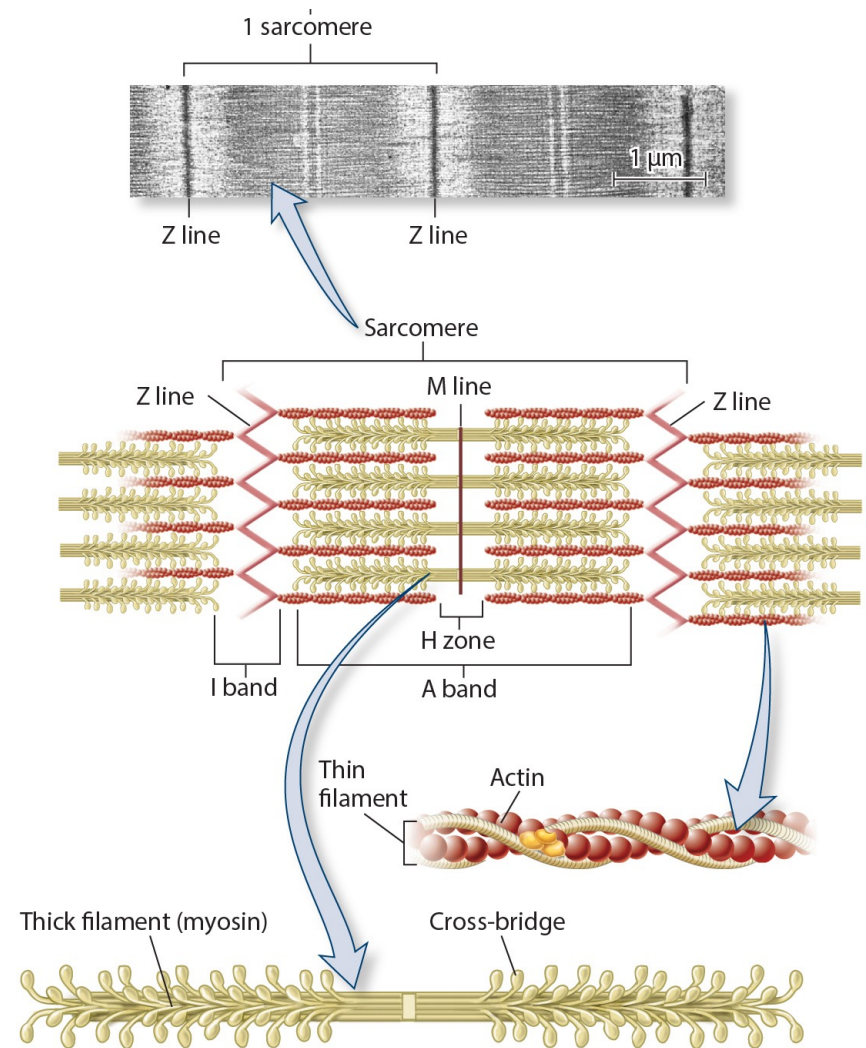
I band – contains portions of thin filaments that do not overlap thick filaments

H zone – narrow region in center of A band, space between the 2 sets of thin filaments

M line – in center of H zone, proteins that link central regions of adjacent thick filaments

Spaces between overlapping thick and thin filaments are bridged by **cross-bridges**

- **Myosin molecules extending from surface of thick filaments toward thin filaments**

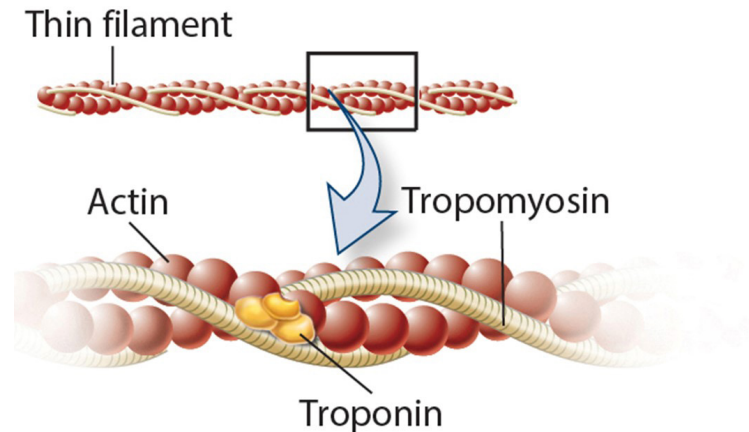


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Filament structure

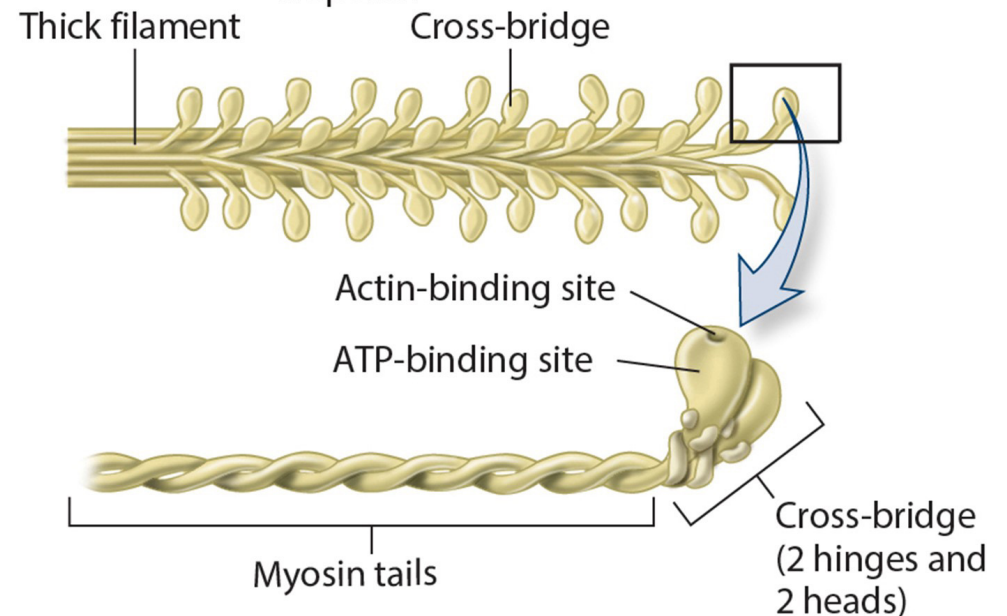
Thin filaments

- **Actin** molecules form **two intertwined helical chains**
- Chains are closely **associated with** two proteins – **tropomyosin** and **troponin** – important in regulating contraction



Myosin (thick filament)

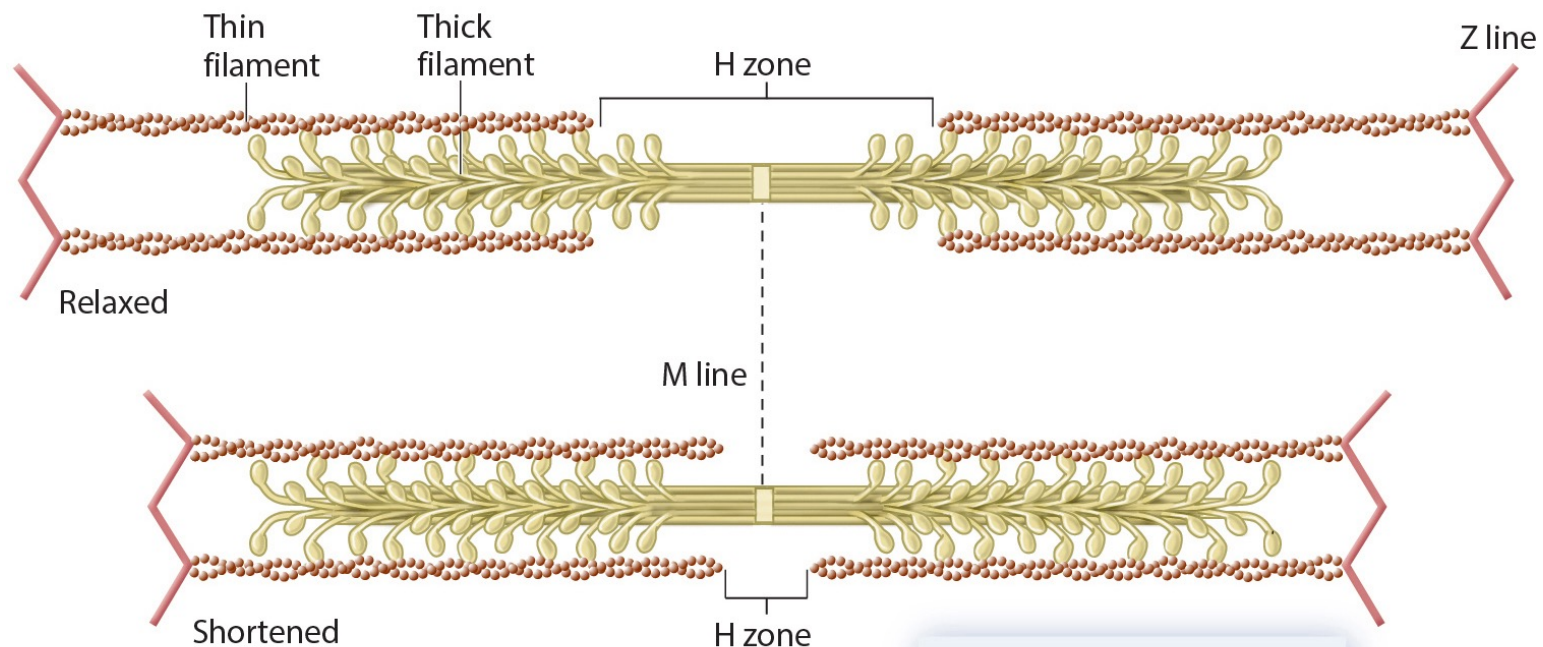
- **Three-domain structure** composed of **two intertwined tails**, **two hinges**, and **two heads**
- Thick filaments are made up of many myosin proteins, with **hinges and heads extending out** to sides and forming **cross-bridges**
- **Each head** contains **binding site** for **actin** and **ATP**



a) Structure of thin and thick filaments

Sliding filament mechanism

- **Sarcomeres shorten** as **thin filaments slide** past stationary thick filaments
- **Myosin cross-bridges** attach to thin filament and force thin filament toward center of sarcomere
- Cross-bridge **repeats motion** as long as stimulation to contract continues



Successive movements of cross-bridges pull the thin filaments toward the M line.

b) Sliding filament mechanism

Sliding filament mechanism



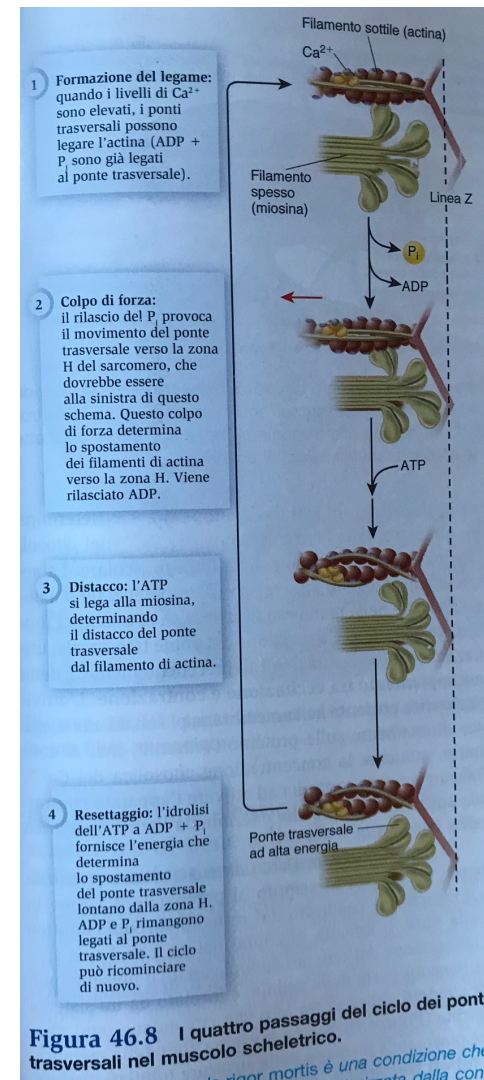
Cross-bridge cycle

Events from the time when the cross-bridge binds to a thin filament and when it is set to repeat the process

- Initiates when Ca^{2+} concentration **exceed critical threshold** cytosol

Four steps

- **Cross-bridges of ADP/Pi-bound Myosins binds to actin** in response to **increase in Ca^{2+}**
- **Cross-bridge and thin filaments move**—“power stroke”- release of ADP and Pi
- **ATP binds** to myosin, and **cross-bridge detaches**
- **Hydrolysis of ATP reenergizes** and **resets the cross-bridge**



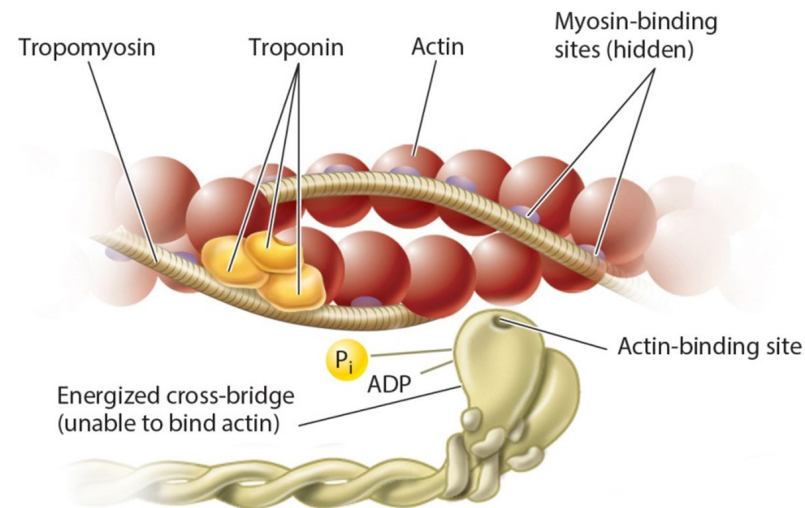
Calcium initiate the process

Tropomyosin

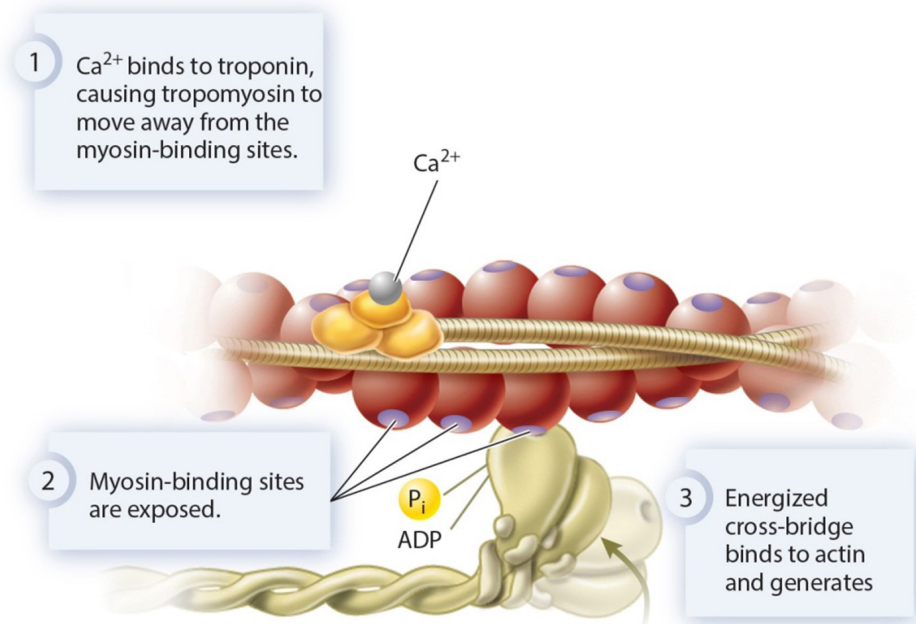
- Composed of two intertwined proteins
- Arranged along length of actin thin filament
- In absence of Ca^{2+} , **covers myosin-binding sites on actin** and prevents cross-bridges' contact with actin

Troponin

- Smaller protein bound to both tropomyosin and actin
- **Binds Ca^{2+} and moves tropomyosin off of myosin-binding site on actin** – permits cross-bridge cycling to occur
- Removal of Ca^{2+} reverses the process → contraction stops



a) Low cytosolic Ca^{2+} , relaxed muscle



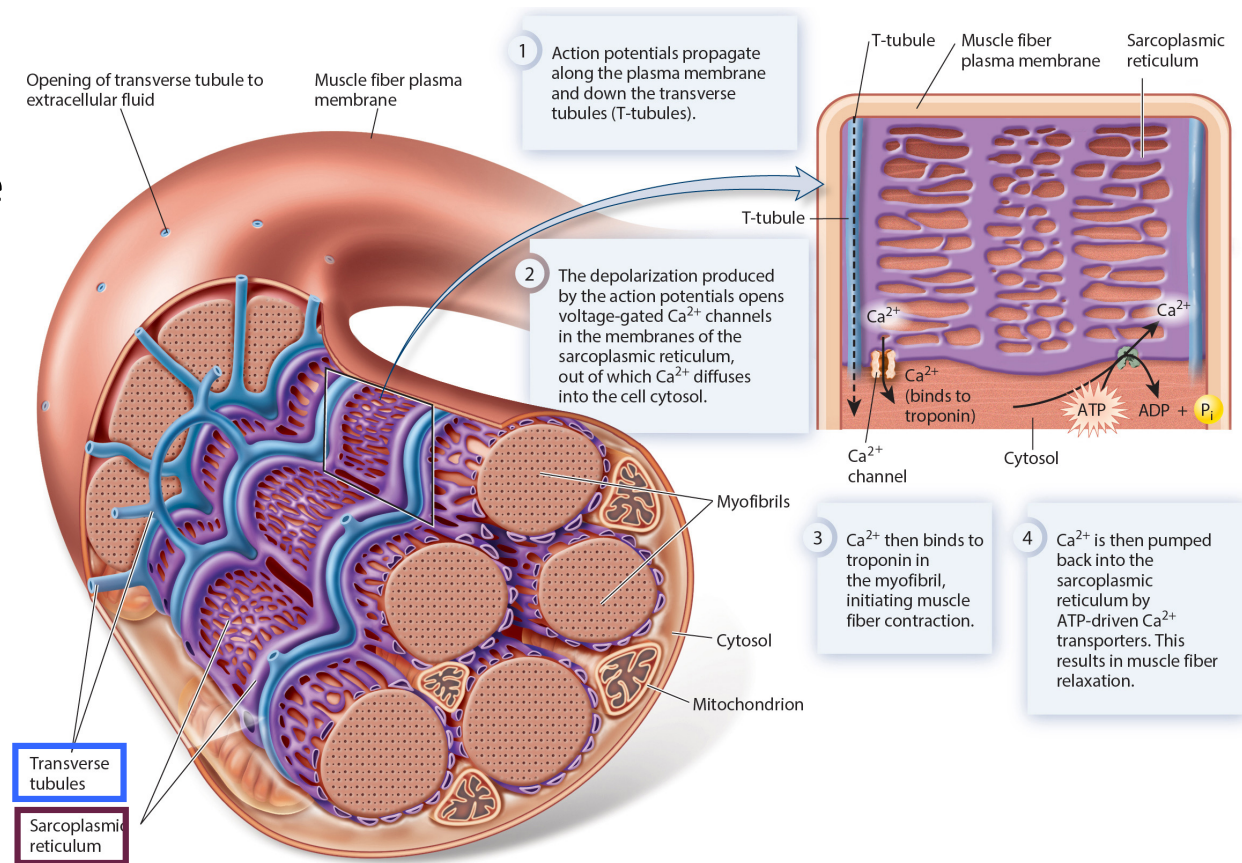
b) High cytosolic Ca^{2+} , activated muscle

Excitation-contraction coupling

Muscle fibers are capable of propagating **action potentials**

Causes rise in cytosolic Ca^{2+} released from **sarcoplasmic reticulum**

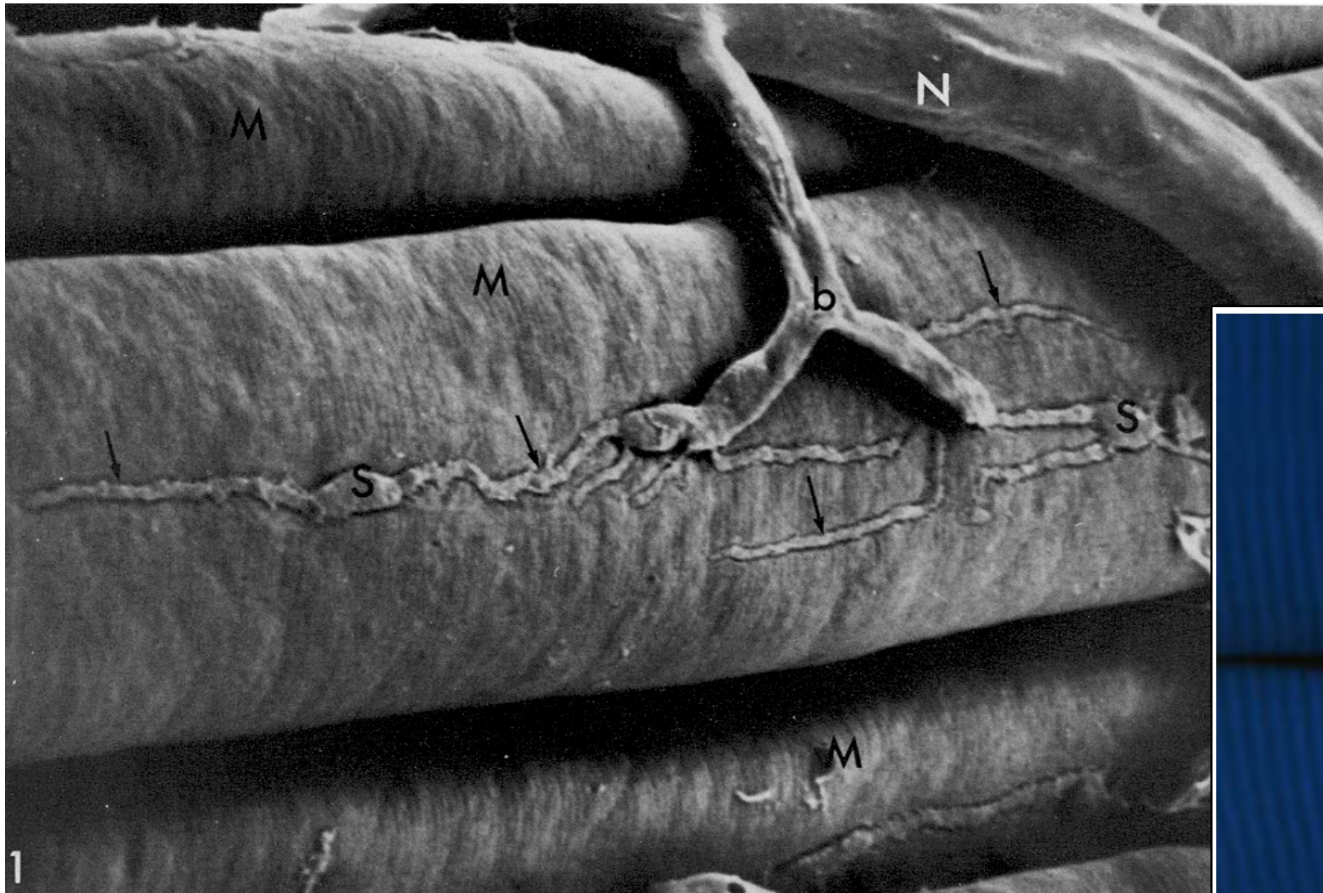
- Transverse or **T-tubules** are **invaginations of plasma membrane** that **conduct the action potential from the outer surface** of muscle fiber to sarcoplasmic reticulum
- Depolarization **opens sarcoplasmic reticulum Ca channels**
- **Rise of cytoplasmic Ca** triggers contraction
- **Ion pumps return calcium to the sarcoplasmic reticulum**, resulting in muscle relaxation



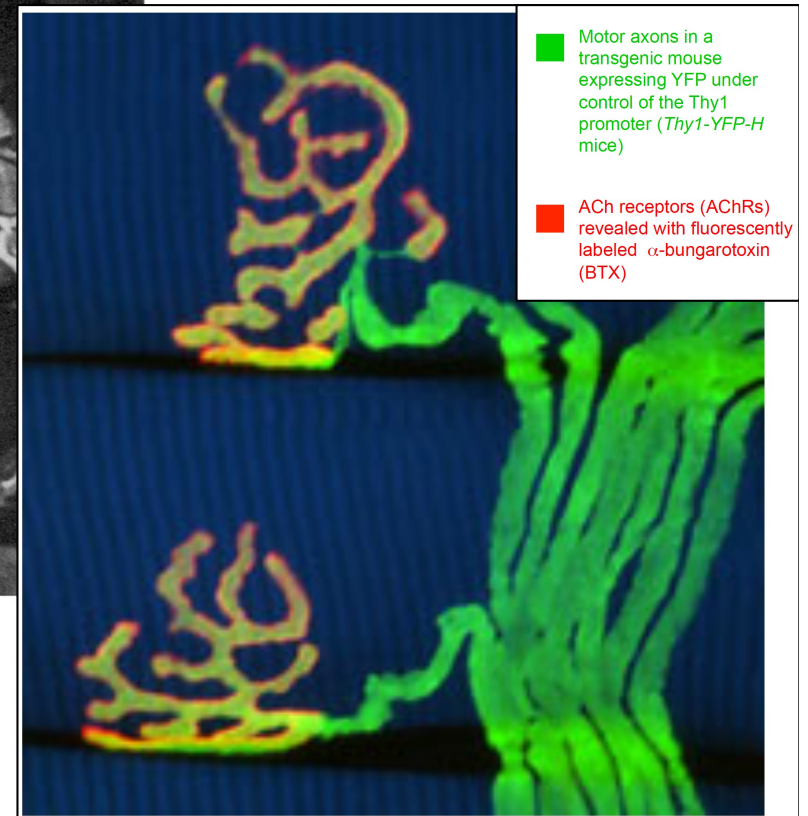
Excitation-contraction coupling



Structure of the NMJ

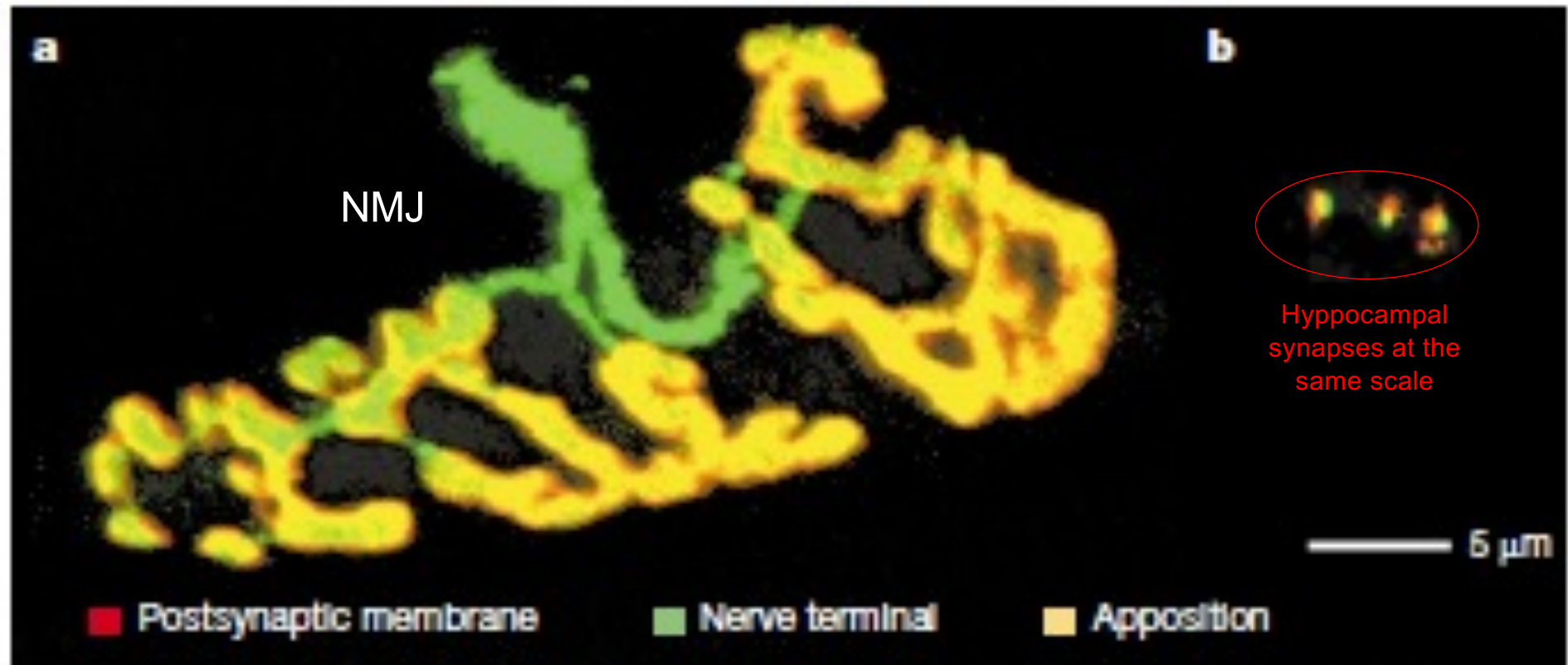


Excitatory synapse
between motorneurons
and muscle fibers



- Neurotransmitter: Acetylcholine
- nicotinic acetylcholine receptors

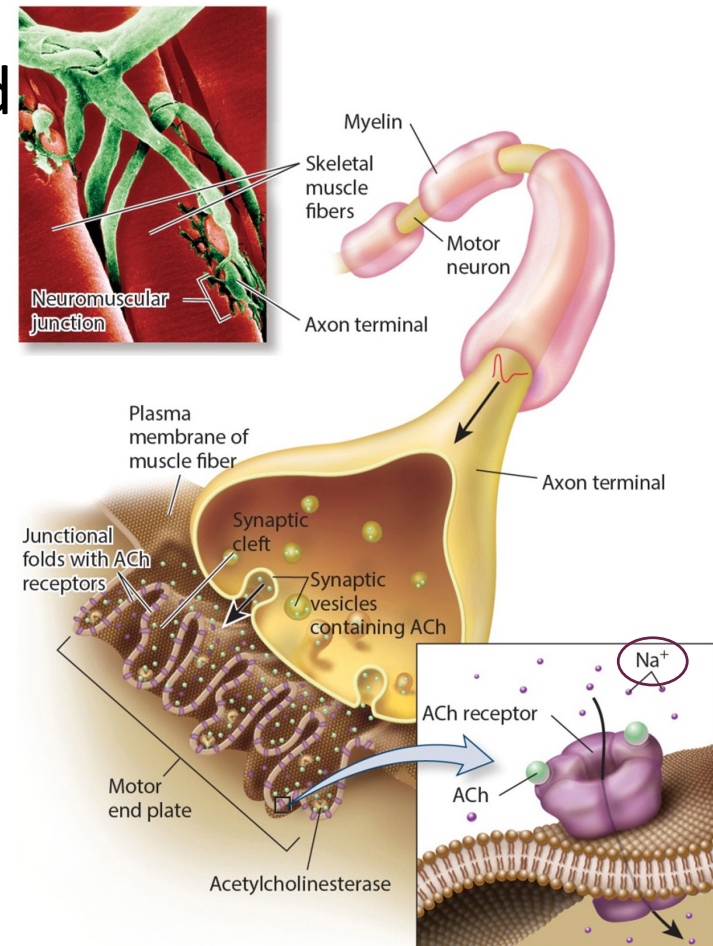
Structure of the NMJ



Sanes and Lichtman 2001

Neuromuscular junction

- Junction of motor neuron's axon and muscle fiber
- **Axon terminals** containing vesicles of **acetylcholine**
- **ACh receptor (nicotinic) on muscle fibers is ligand-gated ion channel**
- **Na⁺ flows into muscle cell** leading to depolarization and action potential propagation to T-tubules
- Overstimulation of muscle fiber prevented by **acetylcholinesterase**



Neuromuscular junction



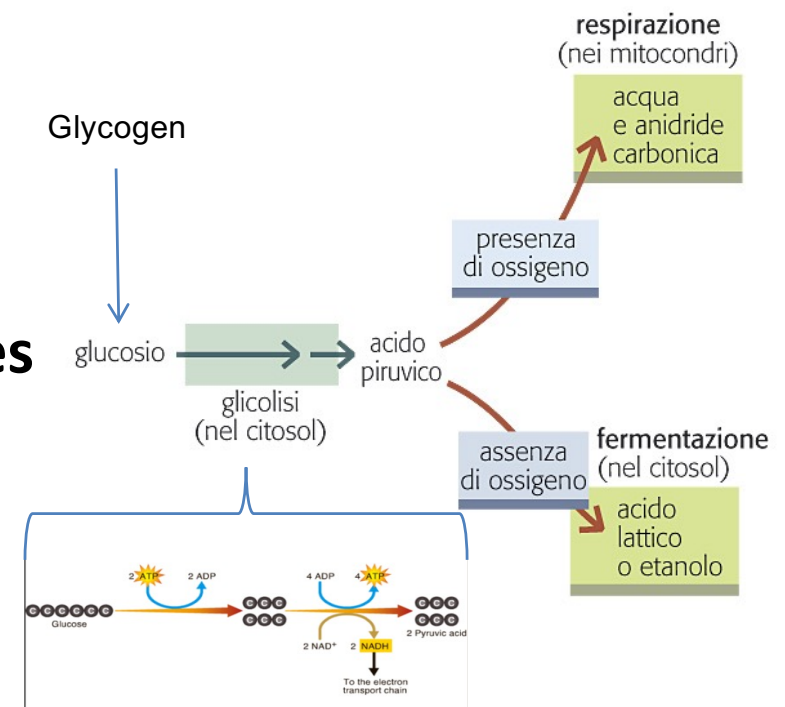
Fibers differ in ATP synthesis ability

Oxidative fibers

- Contain numerous mitochondria and have a high capacity for **oxidative phosphorylation**
- Depends on **blood flow to deliver oxygen** and nutrients for ATP production
- Contain large amounts of **myoglobin** as an **intracellular reservoir of oxygen**

Glycolytic fibers

- **Few mitochondria** but a high concentration of **glycolytic enzymes** and **large stores of glycogen**
- Contain **little myoglobin** – makes them pale or **white**



Muscle Fibers differ in rate of ATP consumption

- Different muscle fibers contain **different forms of myosin** that differ in the **maximal rates at which they can hydrolyze ATP**
- **Slow fibers** – have myosin with a **low ATPase activity**
- **Fast fibers** – contain myosin with **higher ATPase activity**
- Maximal force produced by each is the same, **only rate of cross-bridge cycling varies**

Three major types of fibers

Slow-oxidative fibers

- **Low rates of myosin ATP hydrolysis but makes large amounts of ATP**
- **Used for prolonged activity**

Fast-oxidative fibers

- **High rates of myosin activity, makes large amounts of ATP**
- **Suited for rapid actions**

Fast-glycolytic fibers

- **Very high rates of myosin activity but cannot make as much ATP**
- **Suited for very rapid, intense actions but fatigue quickly**

Table 45.1

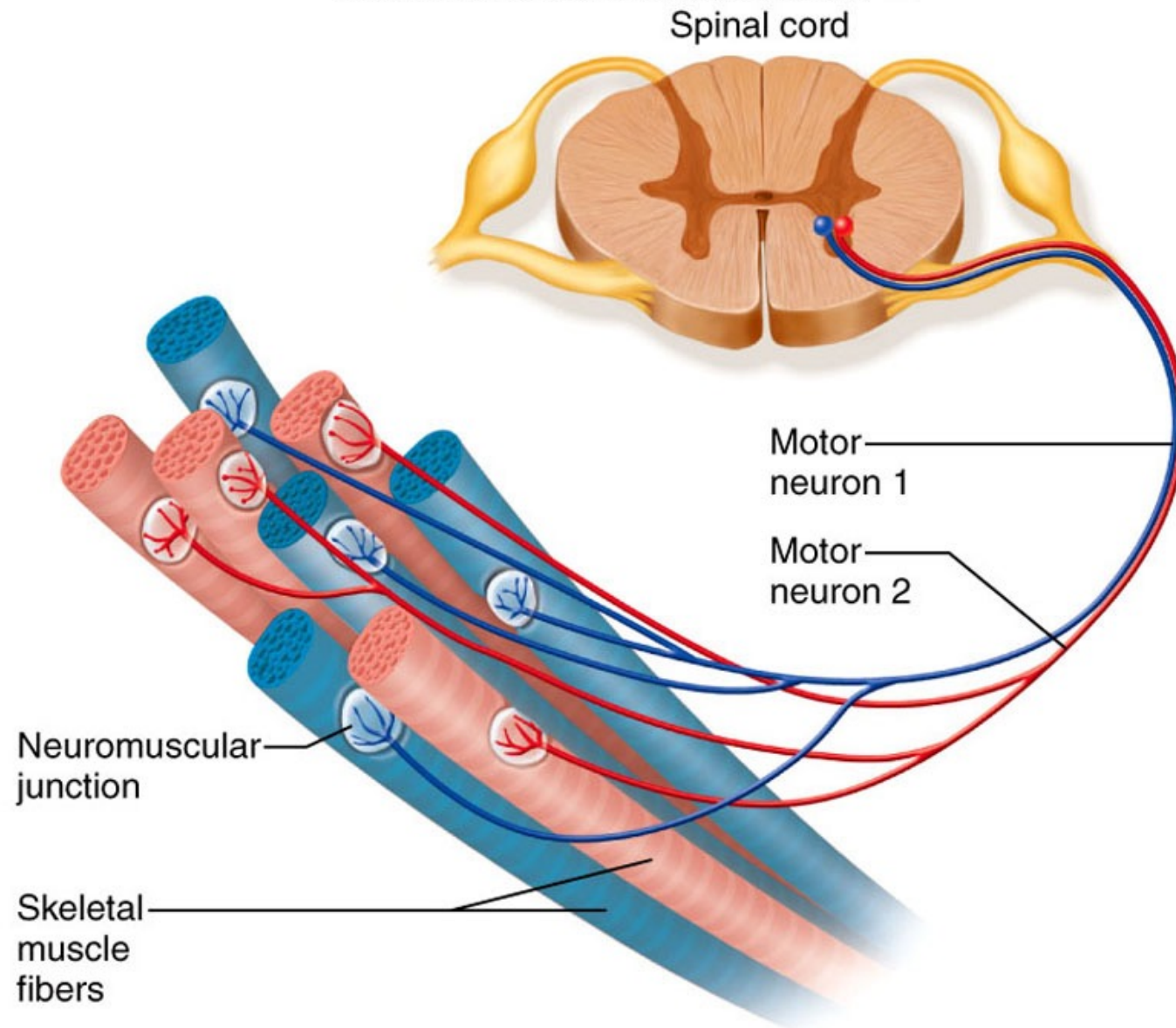
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Table 45.1 Characteristics of the Three Types of Skeletal Muscle Fibers

	Slow-oxidative	Fast-oxidative	Fast-glycolytic
Primary source of ATP production	Oxidative phosphorylation	Oxidative phosphorylation	Glycolysis
Mitochondria	Many	Many	Few
Blood supply	High	High	Moderate
Myoglobin content	High (red)	High (red)	Low (white)
Rate of fatigue	Slow	Intermediate	Fast
Myosin ATPase activity	Low	High	High
Rate of contraction	Slow	Fast	Fast

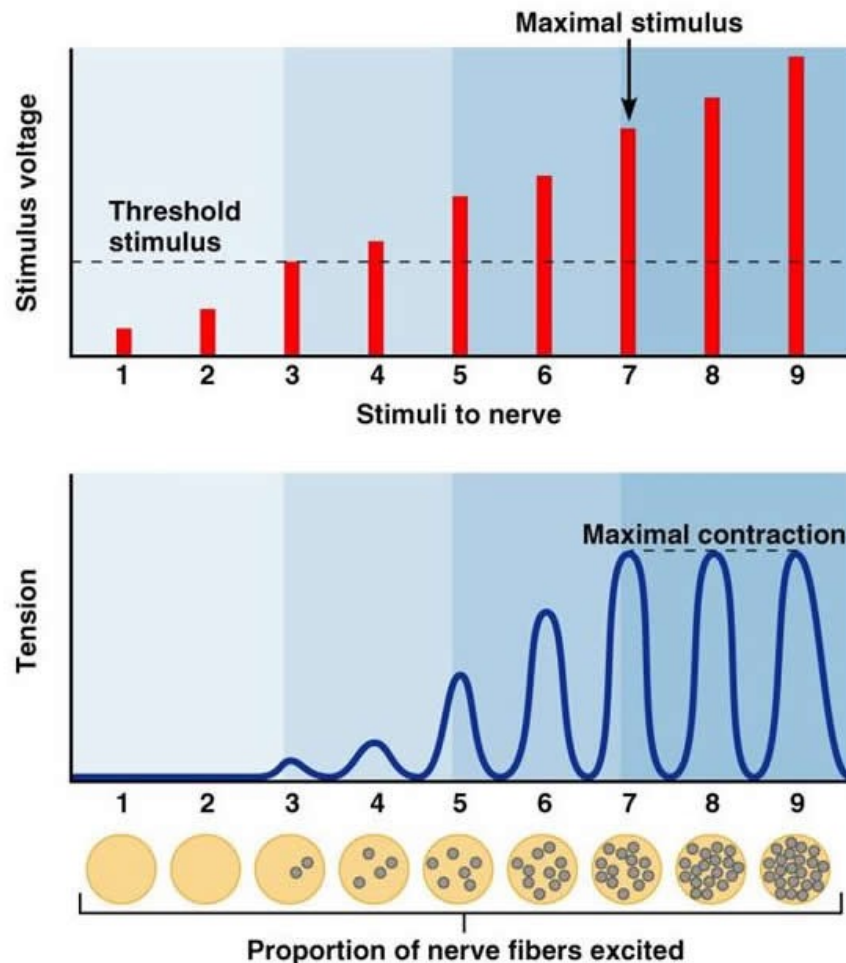
Motor units

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Muscle contraction and motor unit

- When a **motor unit** is activated, all of its fibers contract.
- In vertebrates, the **force of a muscle contraction** is controlled by the **number of activated motor units**.



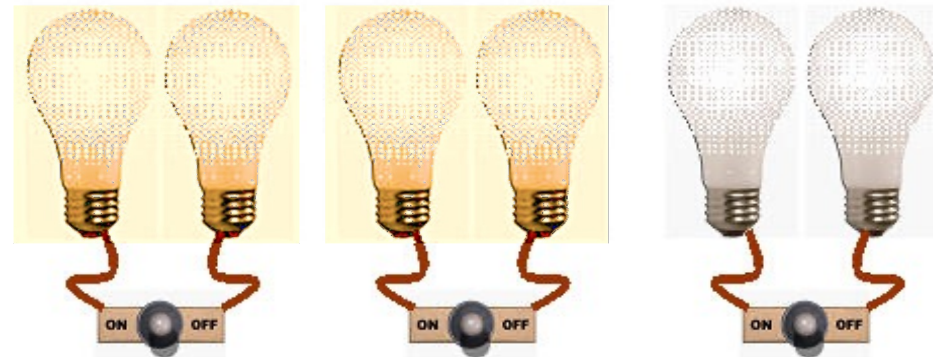
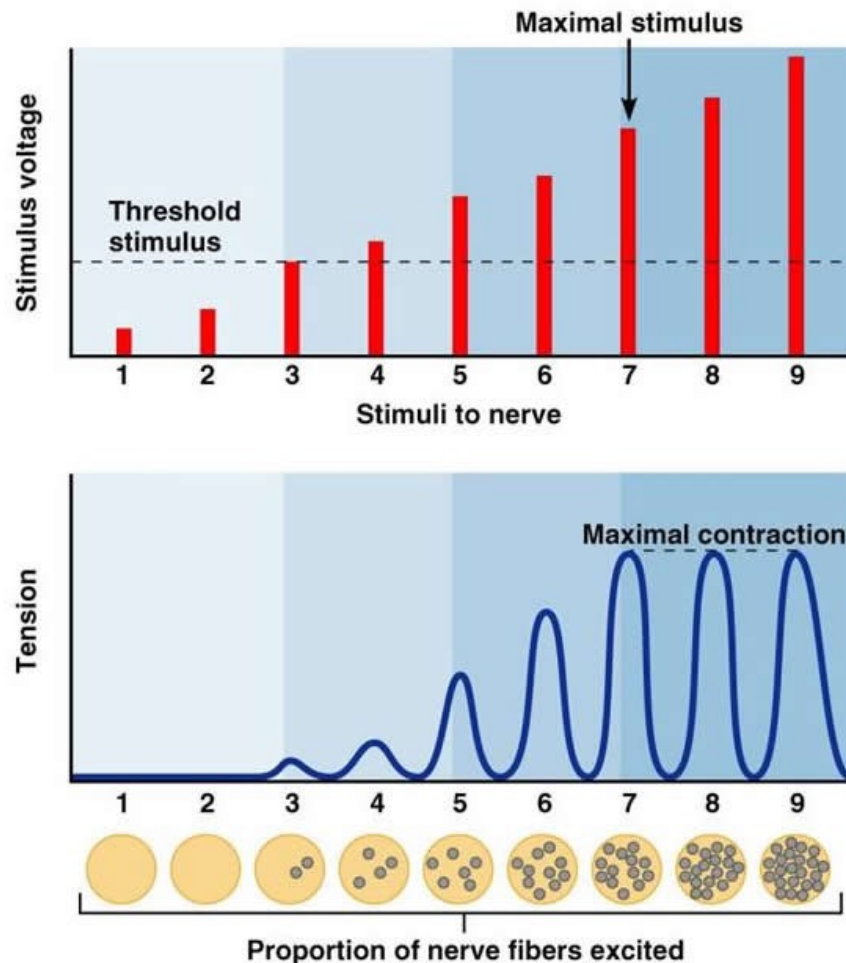
Motor unit analogy

If you want more light,
you turn on more than one lighting unit

If you want more muscle tension,
you turn on more than one motor unit

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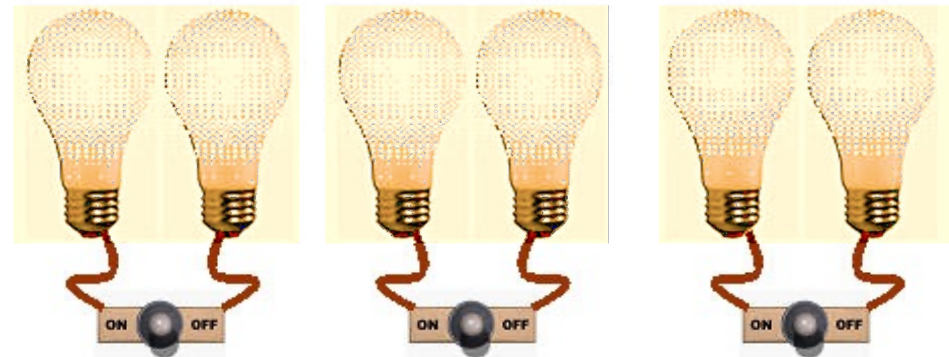
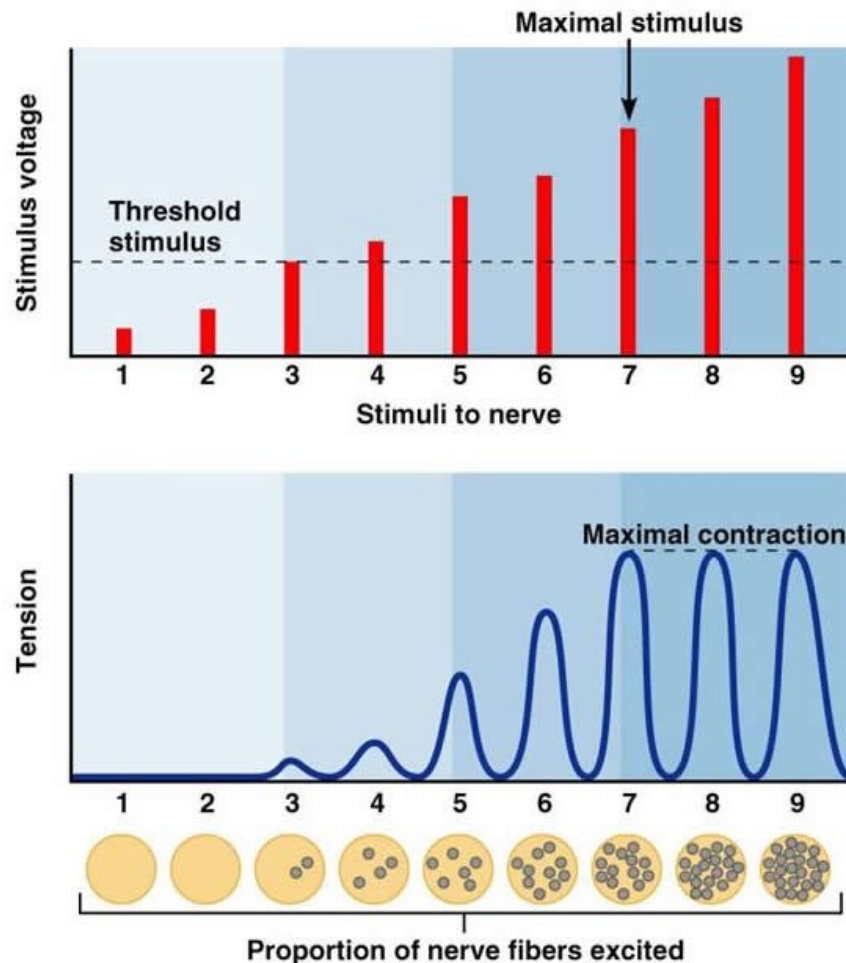
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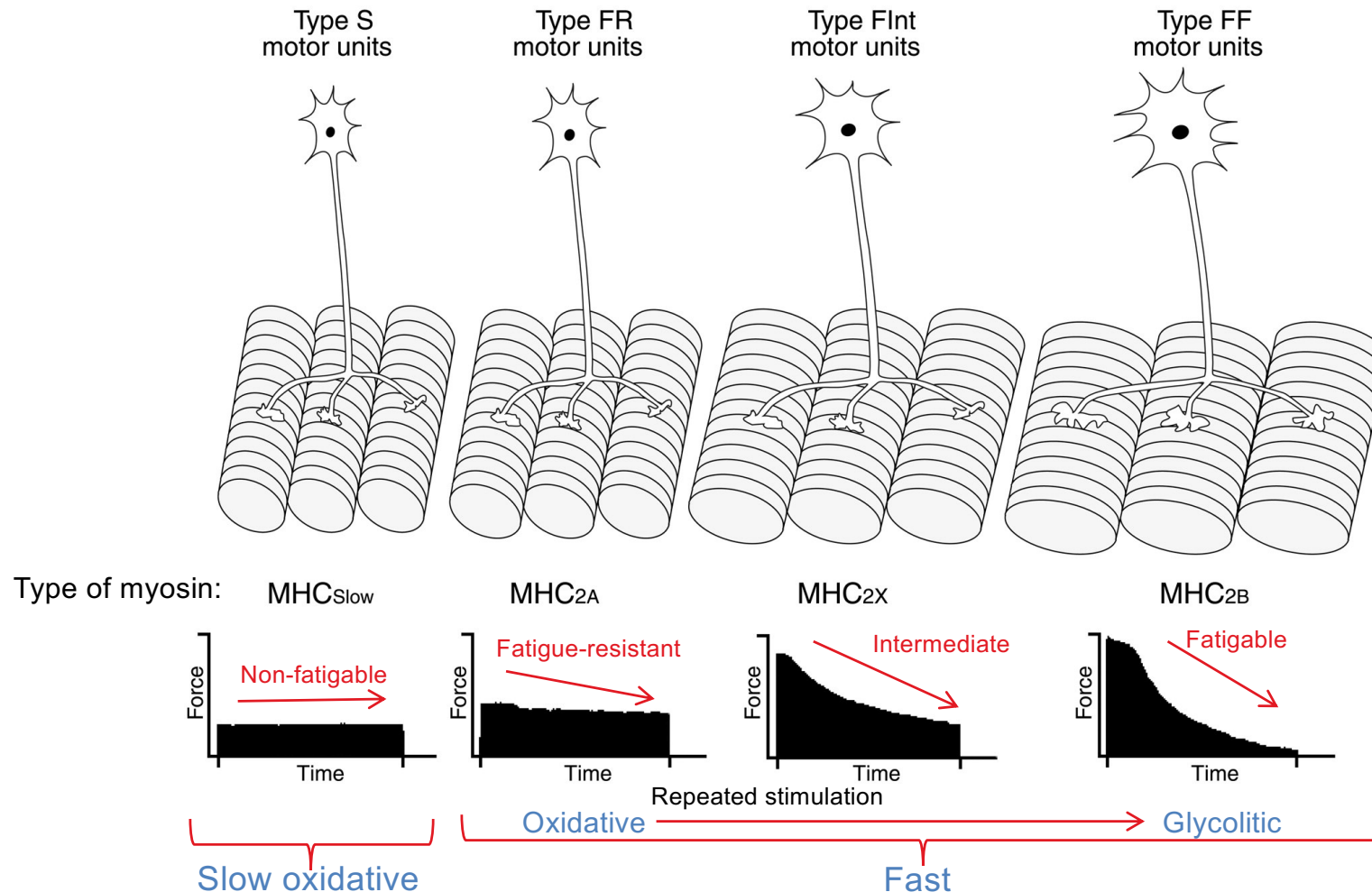
Motor unit analogy

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Motor units-muscle fiber type

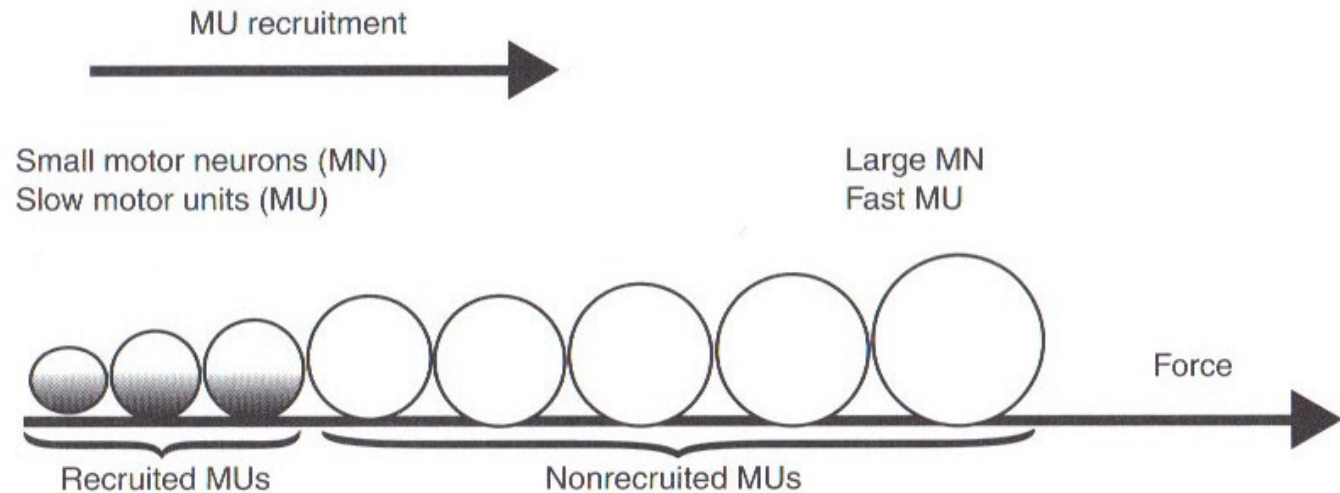
- All muscle fibers in a motor unit are of the **same fiber type**



- The functional distinctions between motor units also explain some structural differences among muscle groups. For example:
 - in the **soleus** (a muscle important for posture; mostly slow units) has an innervation ratio of **180 muscle fibers for each motor neuron**
 - the **gastrocnemius**, has an **innervation ratio of 1000–2000 muscle fibers** per motor neuron → generate forces needed for sudden changes in position.
 - extraocular muscle** motor units are extremely small (with an **innervation ratio of only 3**) and have a high proportion of muscle fibers capable of contracting with maximal velocity to allow **rapid, precise movements**

Henneman's size principle

- Motor units (MU) recruitment is **sequential** in a **size dependent** way
- **Smaller motor units (less force)** innervated by smaller neurons are **recruited first**, larger motor units later.



- In individual muscles, **size and force** of the **motor unit** are **correlating** with the **type of muscle fibers** that are involved

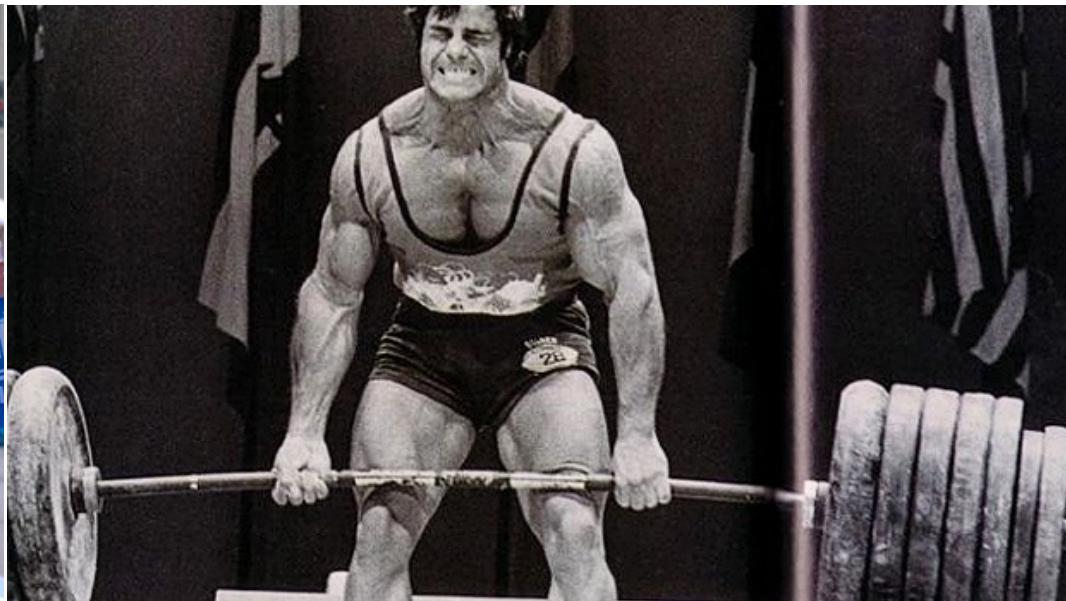
Motor Unit (MU) type	Type of muscle fibers	Size of the MU (n. fibers)	Force	Excitability
S-type	Slow/unfatigable	Small	Small	High
FR-type	Fast/fatigue-resistant	Large	Large	Average
FF	Fast/fatigable	Very large	Very large	Low

Exercise

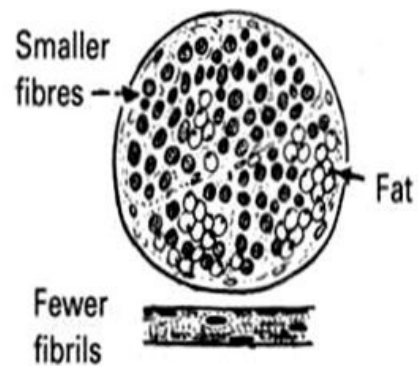
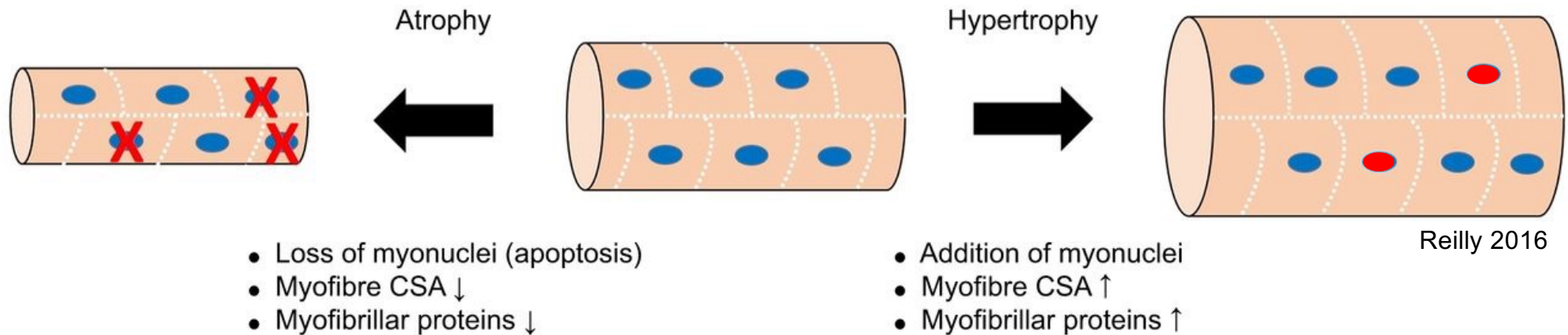
Increased amounts of exercise can produce an **increase in the size of muscle fibers** and/or their **capacity for ATP production**

Adaptive response depends on type of exercise:

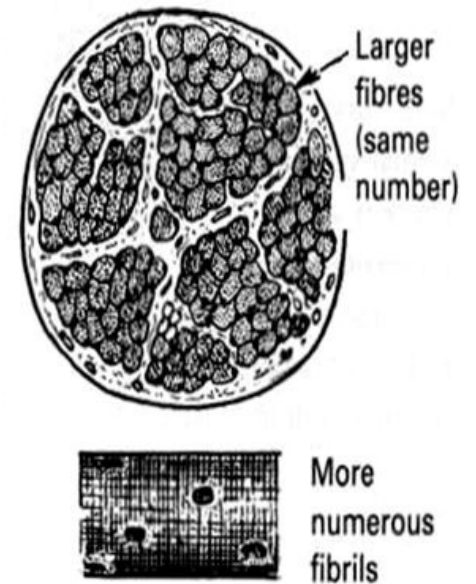
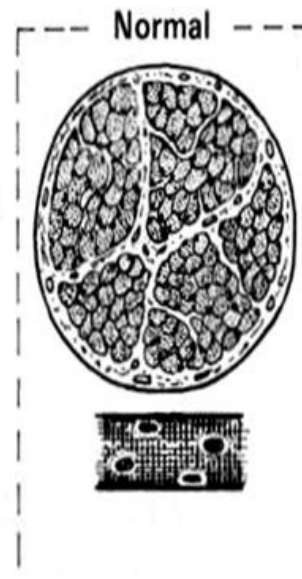
- **Aerobic exercise** increases **number of mitochondria** and **blood vessels** surrounding fibers
- **Strength exercise** increases **size of fibers**



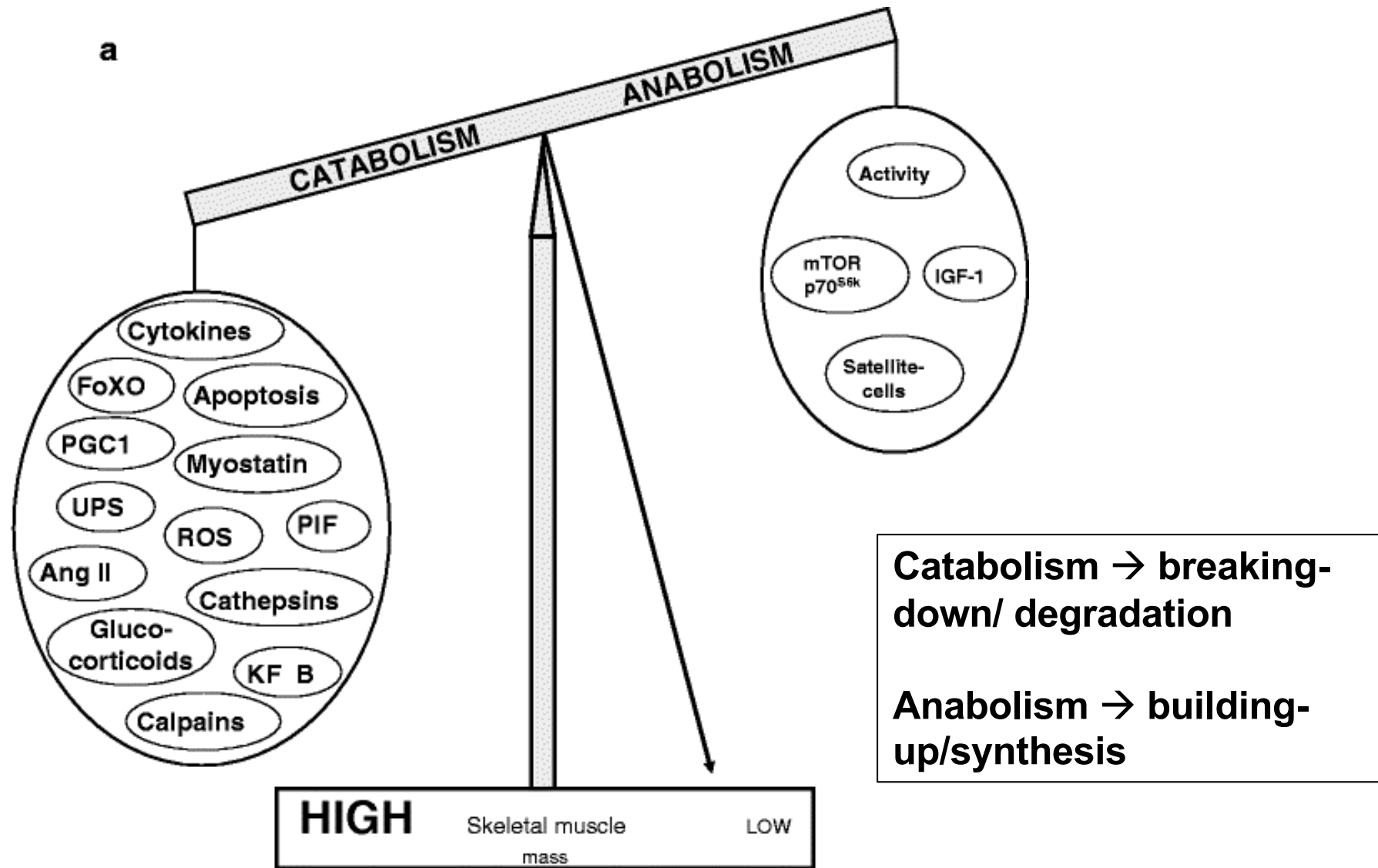
Skeletal muscle remodeling



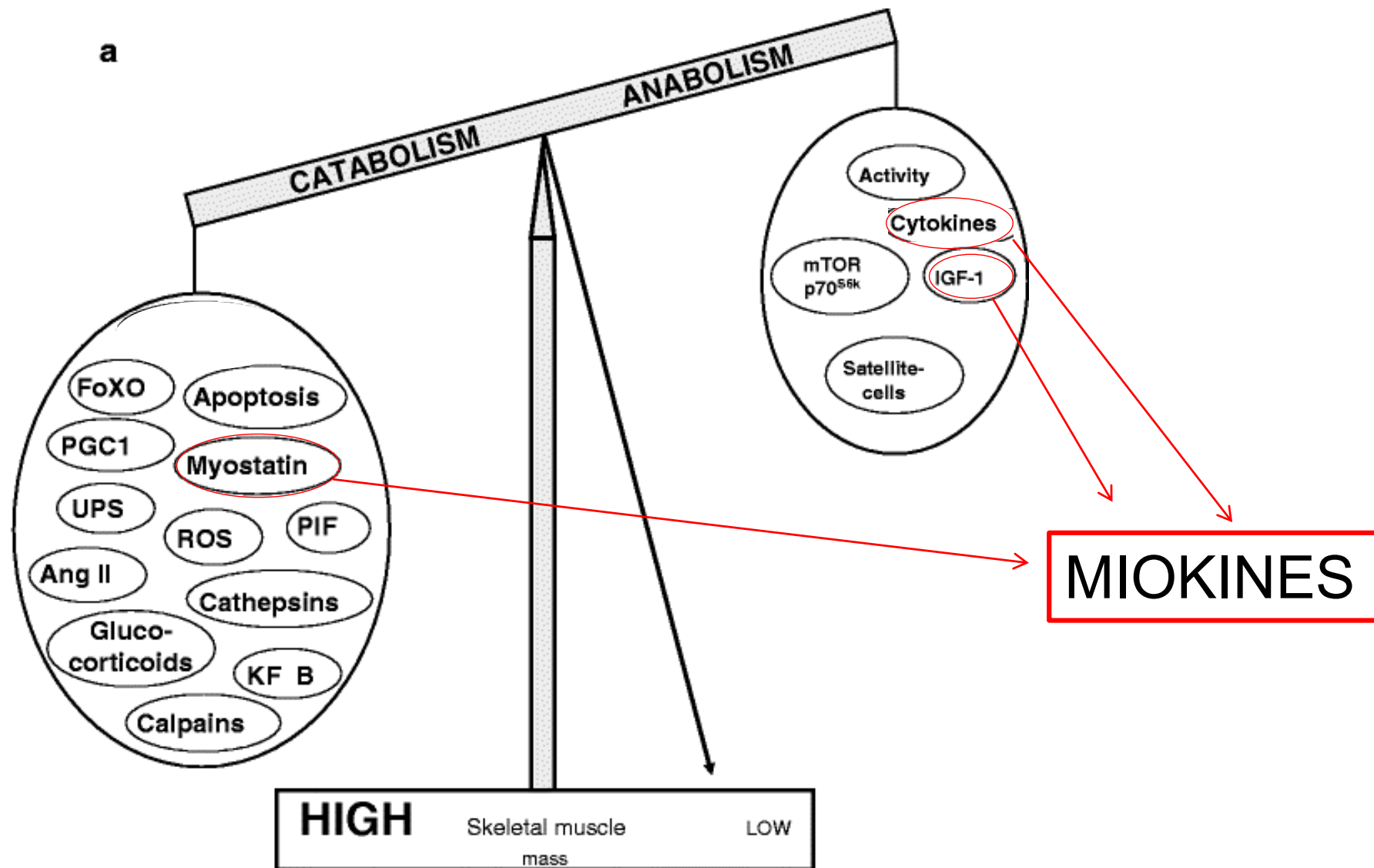
Atrophy may be partially masked by increased fat in the septal tissues.



A balance between catabolism and anabolism

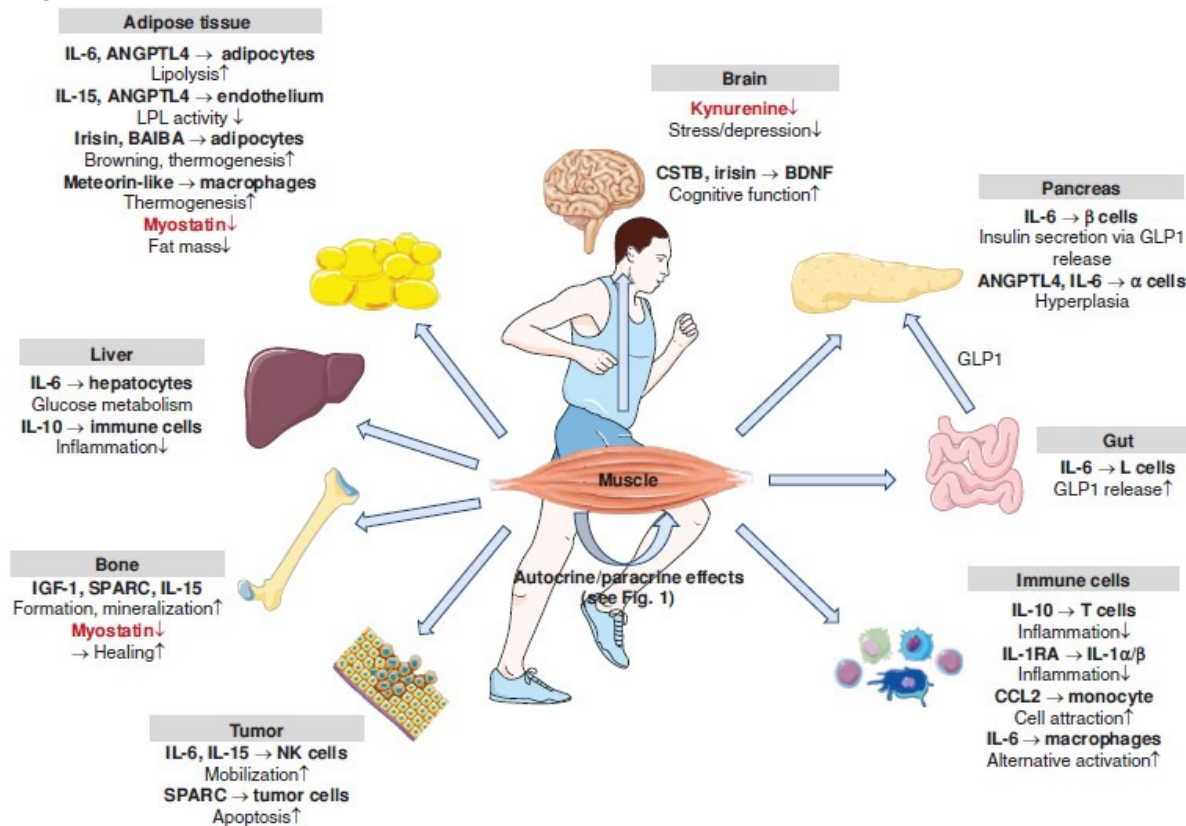


Myokines modulate catabolism and anabolism

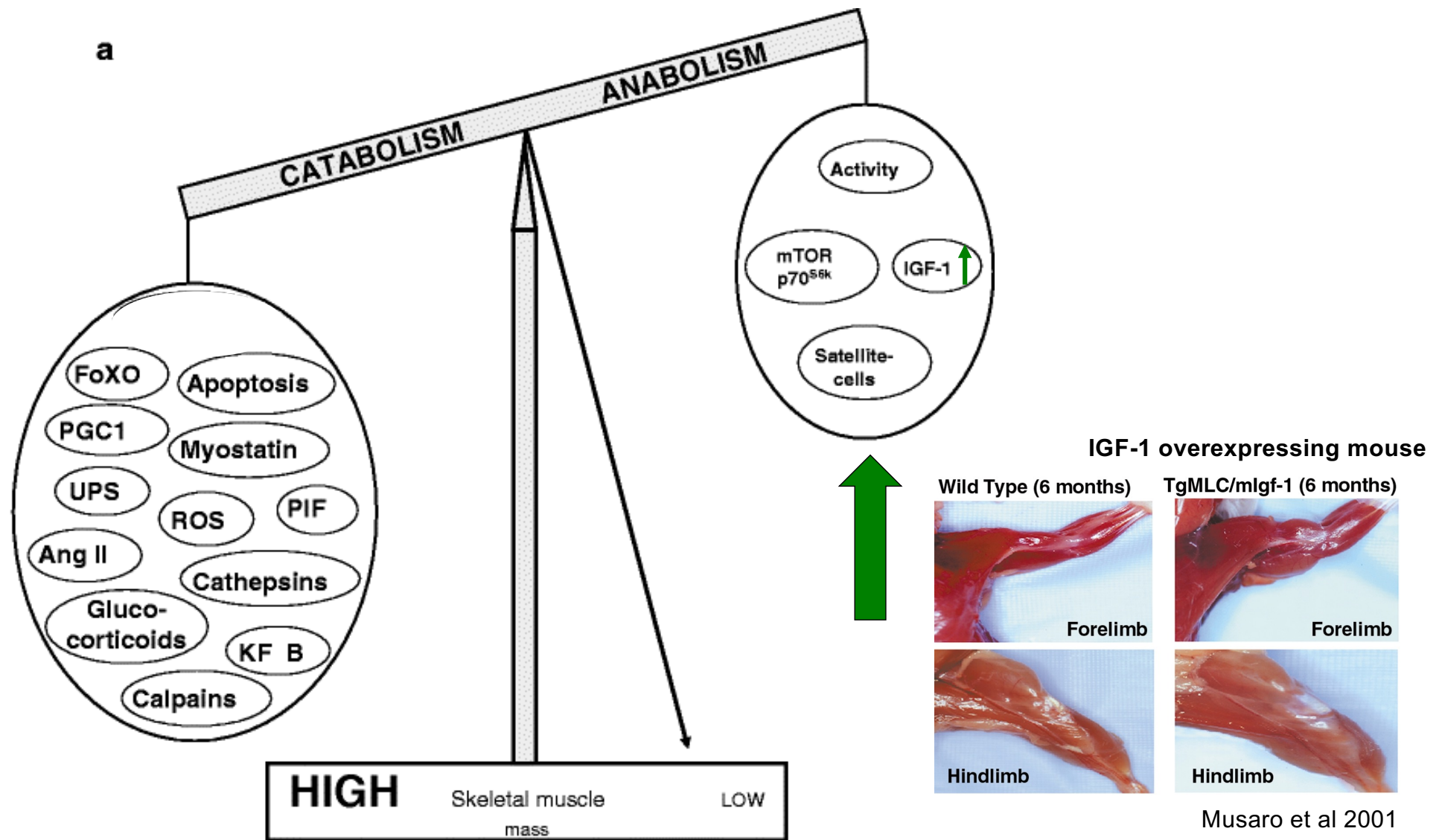


Miokines and systemic adaptation to exercise

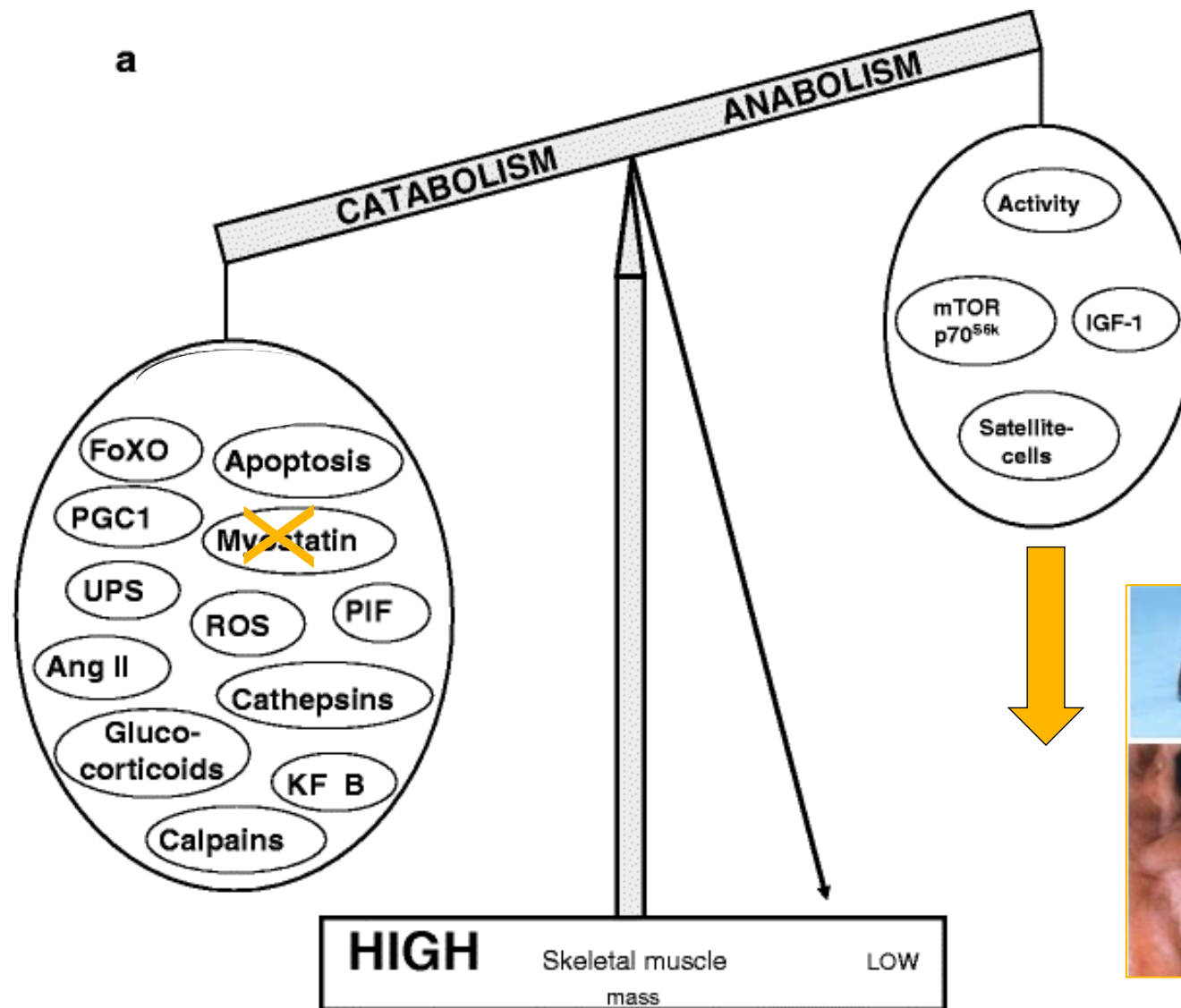
- **Myokines**
 - are released by skeletal muscle
 - may be modulated by exercise or exercise training
 - may be released in the circulation and control different tissues → endocrine function
 - May act locally in the skeletal muscle → para/autocrine function



IGF-1 versus Myostatin



IGF1 versus Myostatin



Myostatin KO mice



Myostatin plays a conserved role



Whippet



Shelton 2007

Belgian blue cattle



Myostatin plays a conserved role

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RESEARCH ARTICLE

Muscle growth in teleost fish is regulated by factors utilizing the activin II B receptor

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Transgenic trout expressing an **inactive truncated form of the myostatin receptor** *acvr2b-2a* (*acvr2b^Δ*) in muscle tissue



Myostatin plays a conserved role

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RESEARCH ARTICLE

Muscle growth in teleost fish is regulated by factors utilizing the activin II B receptor

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Transgenic trout expressing an inactive truncated form of the *acvr2b-2a* (*acvr2b^Δ*) in muscle tissue



Michael Fred Phelps



Myostatin mutations and human muscle mass

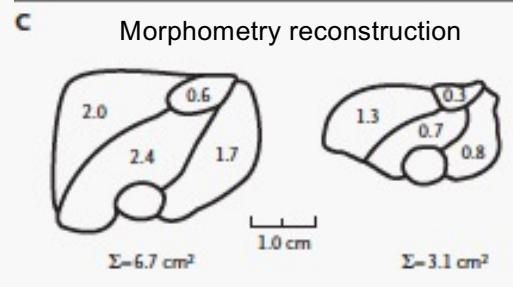
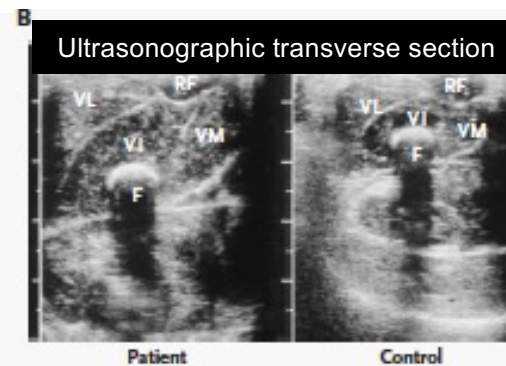
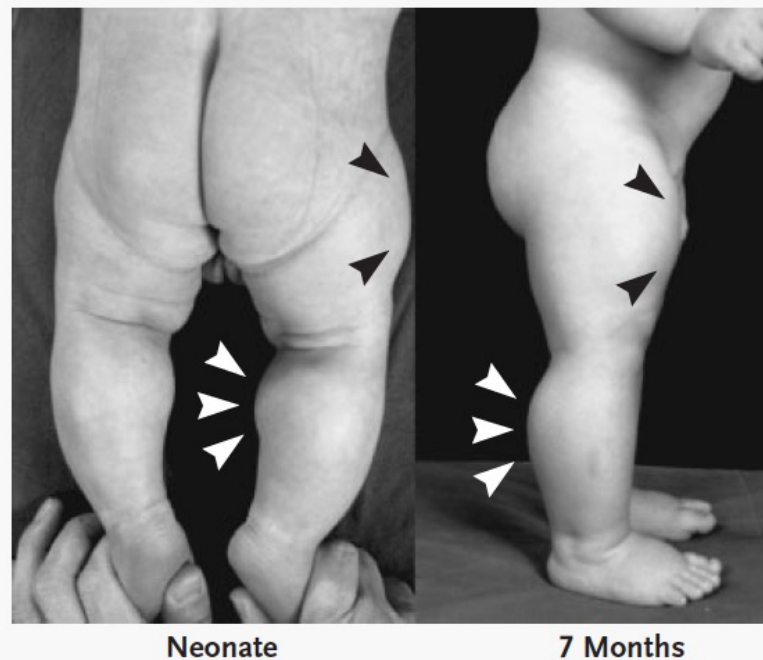
The NEW ENGLAND JOURNAL of MEDICINE

BRIEF REPORT

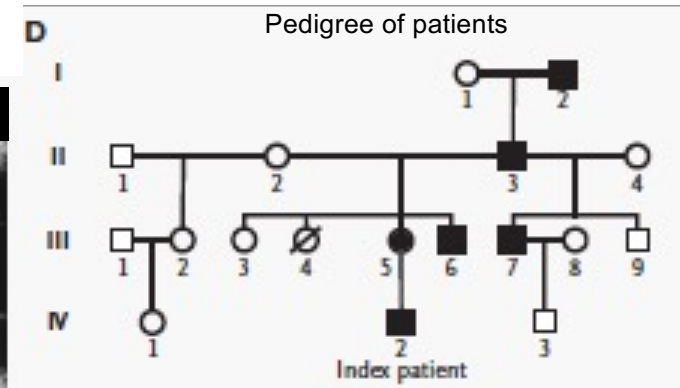
Myostatin Mutation Associated with Gross Muscle Hypertrophy in a Child

Markus Schuelke, M.D., Kathryn R. Wagner, M.D., Ph.D., Leslie E. Stolz, Ph.D., Christoph Hübner, M.D., Thomas Riebel, M.D., Wolfgang Kömen, M.D., Thomas Braun, M.D., Ph.D., James F. Tobin, Ph.D., and Se-Jin Lee, M.D., Ph.D.

N ENGL J MED 350:26 WWW.NEJM.ORG JUNE 24, 2004



Tight muscles



- **Mother** was a **professional athlete**
- Several **family members** have been reported to be **unusually strong** (II-3 was a construction worker able to unload curbstones by hand)
- **Heterozygous mis-splicing of the myostatin precursor mRNA** in the patient, due to splice-site mutation leading to a **truncated protein**

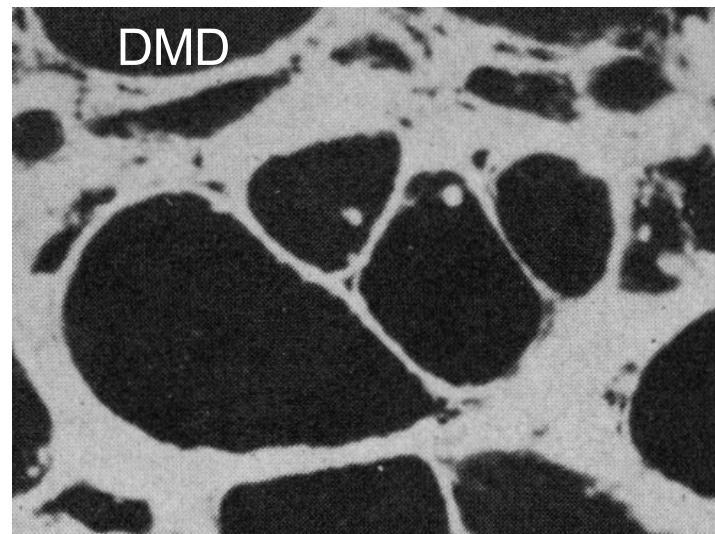
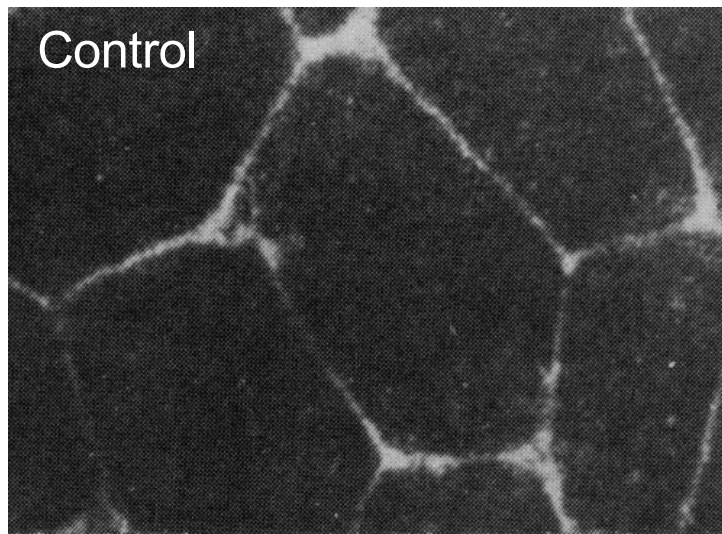
Impact on Public Health: skeletal muscle

Muscular dystrophy

Group of diseases affecting 1 in 3500 American males (less common in females)

Progressive degeneration of skeletal and cardiac muscles, ultimately leading to death

Most common form is **Duchenne muscular dystrophy (DMD)** – sex-linked recessive disorder

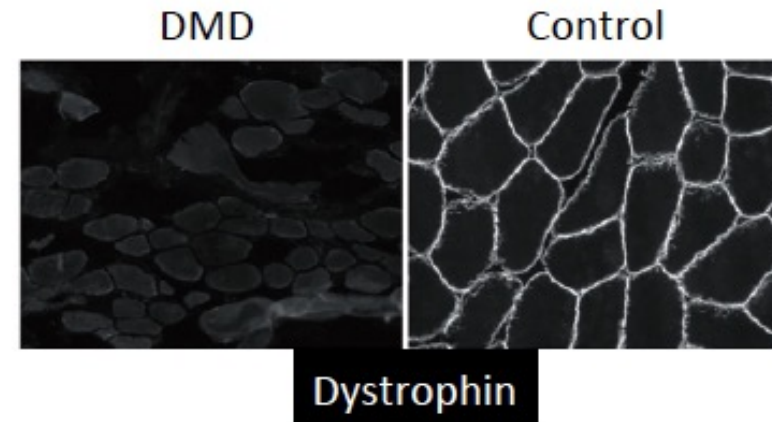
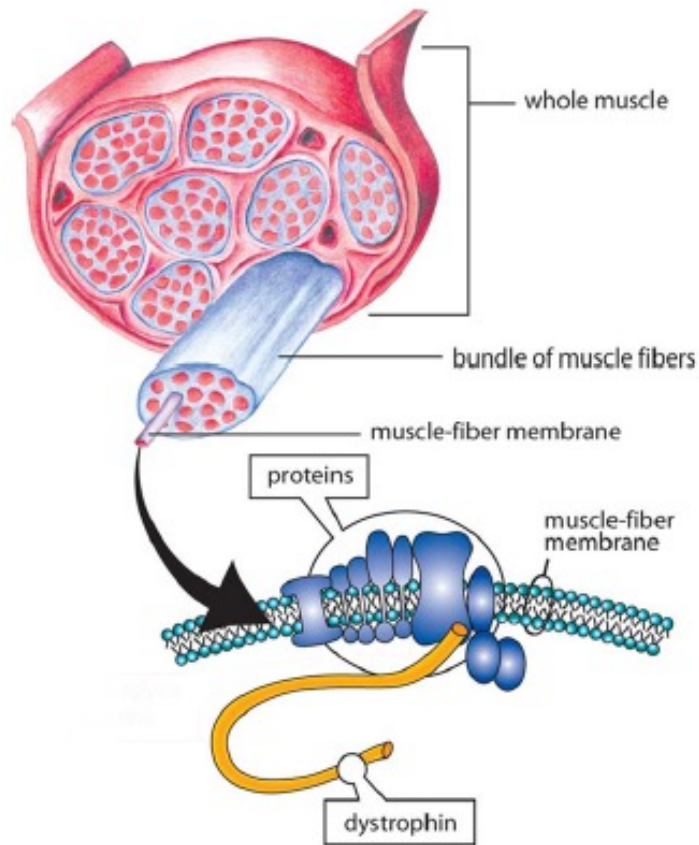


Collagen 1

Modified from
Foidart et al
1981

Duchenne Muscular Dystrophy (DMD) and Dystrophin

- Mutation in the Dystrophin gene on the X chromosome (Hoffman et al. Cell. 1987)



Dystrophin deficiency



Abnormal mechanical properties

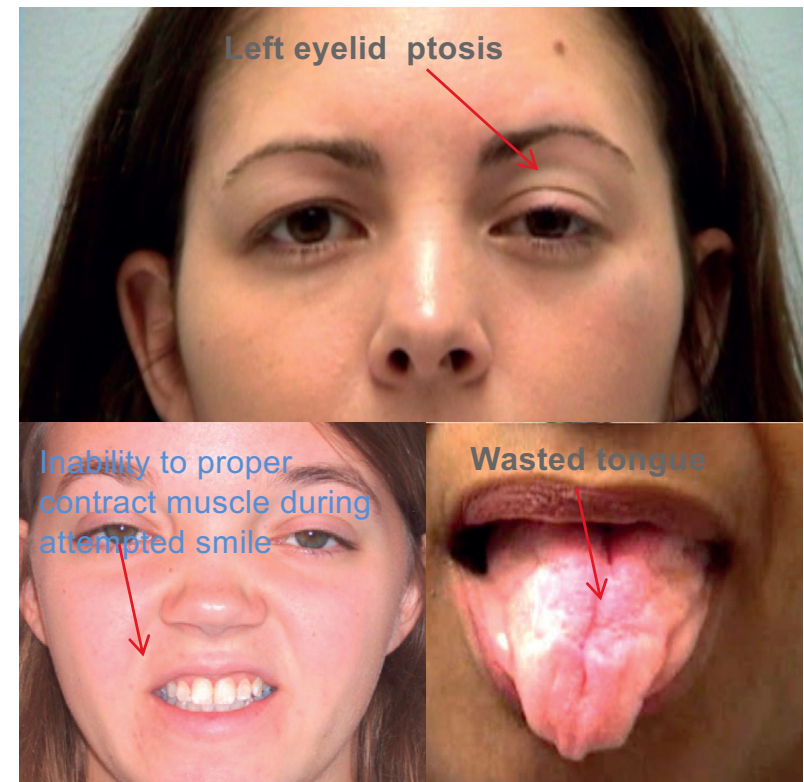


Fibers fragility

- Patients **lack dystrophin**, a protein thought to maintain structural integrity of plasma membrane of muscle fibers

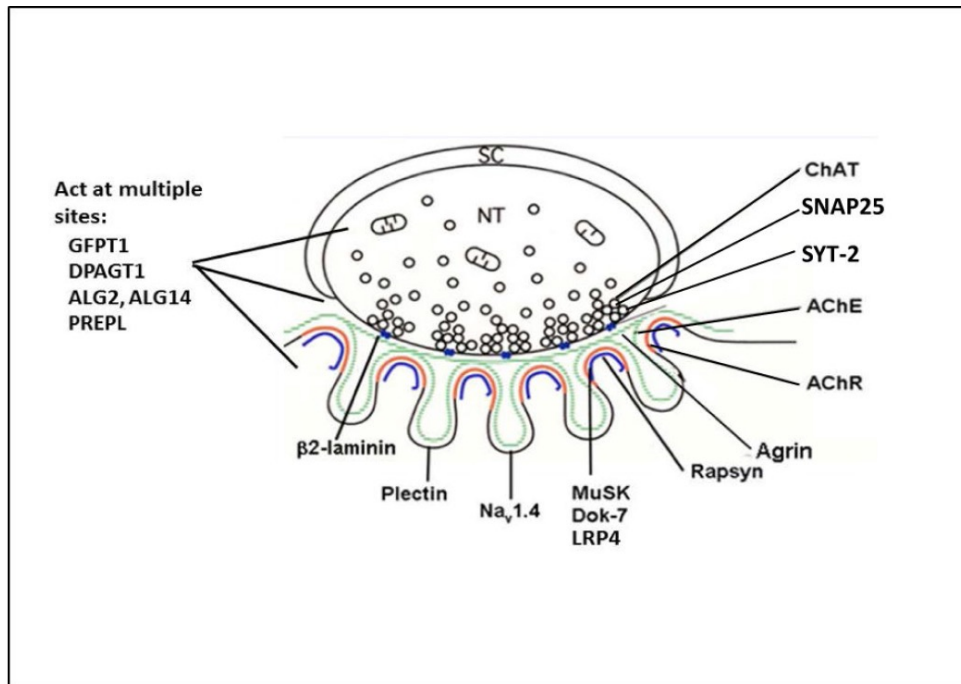
Myasthenia gravis: a NMJ disease

- **Prevalence** of 40-180/million people
- The distribution of **muscle weakness** is **fluctuating** not only **between patients** but also during the **progression of the disease** in each patient
- Three main muscle groups are affected:
 - **extraocular** muscles
 - **Bulbar (oro-pharyngeal)** muscles
 - **Limb and axial muscles**
- In the majority of patients present **initially ocular weakness**
- **Bulbar (oro-pharyngeal)** muscle-dependent **difficulty in swallowing** and **respiratory insufficiency** are **life-threatening**
- **Muscle weakness** increase the **risk of respiratory infections** and **osteoporosis**
- In the majority of **MG patients (~85%) anti-AchR IgGs** have been found, in the remaining 15% autoantibodies against other NMJ components
→ **autoimmune disease**



Congenital myasthenic syndromes

- Extremely rare diseases: ~ **3 cases / million**
- The majority of mutations are **autosomal recessive**
- Are classified in **presynaptic, synaptic and postsynaptic**



Engel 2015

Table 1. Congenital myasthenic syndromes.

Anatomical site	Defect
Presynaptic	<u>Defective acetylcholine synthesis or packaging</u> (episodic apnea, "familial infantile myasthenia," abnormal choline acetyltransferase) Reduced quantal release with/without synaptic vesicles <u>Deficient acetylcholine release</u> by nerve impulses Lambert-Eaton-like syndrome
Synaptic	<u>Endplate acetylcholinesterase deficiency</u>
Postsynaptic	<u>Decreased response to acetylcholine:</u> acetylcholine receptor subunit mutations; rapsyn mutations (receptor clustering); fast-channel syndromes <u>Increased response to acetylcholine:</u> slow-channel syndromes Plectin deficiency with epidermolysis bullosa

Keesey 2004

Impact on Public Health: bone

Osteoporosis

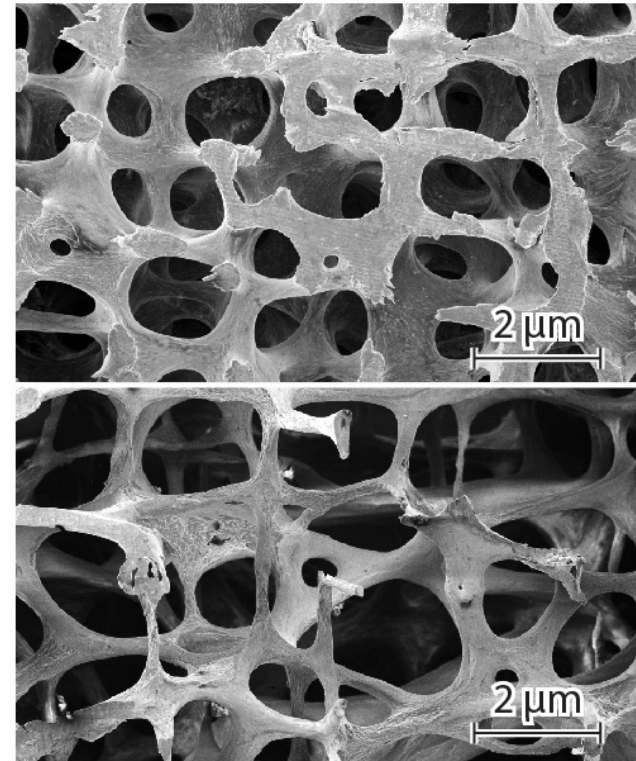
Both mineral and organic portions of bone reduced

Impacts 4 times **more women** than men

Occurs when normal balance between bone formation and breakdown is disrupted

- Caused by **prolonged disuse of muscles** or **hormonal imbalance**

Can be minimized with **adequate calcium** intake, **weight-bearing exercise**, and adequate **dietary Vitamin D**



b) Histologic appearance of normal bone (top) and bone from a person with osteoporosis (bottom)

Chapter 45 Lecture Outline

See separate PowerPoint slides for all
figures and tables pre-inserted into
PowerPoint without notes.

Bibliography

